JustCmul8

REFERENCE GUIDE · OCTOBER 2026

Simulation Playbook

Eighteen step-by-step builds across six domains, from a single coffee counter to conveyor lines, machine bottlenecks and network load, plus thirty industry use cases for pitching.

18 guides | 6 domains | 30 use cases

For	Date	Built against
Builders, testers, sales	26 October 2026	JustCmul8 @ b1dfd7d

00 CONTENTS

Two parts: build, then pitch

Part A is a set of recipes you can follow click by click; each one ends with the numbers you should see, so it doubles as a test case. Part B lists where each domain earns its keep in industry.

Part A · Step-by-step guides

01	Coffee Shop Counter	PEOPLE & SERVICE	EASY
02	Metro Station Entry	PEOPLE & SERVICE	MEDIUM
03	Hospital OPD Patient Flow	PEOPLE & SERVICE	ADVANCED
04	Fuel Station Pumps	TRAFFIC & VEHICLES	EASY
05	Toll Plaza: FASTag vs Cash	TRAFFIC & VEHICLES	MEDIUM
06	Car Service Centre	TRAFFIC & VEHICLES	ADVANCED
07	Storm-water Retention Tank	LIQUID & MATERIAL	EASY
08	Two Tanks, One Valve	LIQUID & MATERIAL	MEDIUM
09	Batch Reactor with Quality Split	LIQUID & MATERIAL	ADVANCED
10	Single CNC Machine	MANUFACTURING	EASY
11	Two-Machine Line with QC Rework	MANUFACTURING	MEDIUM
12	Welding Bottleneck with Downtime and a Paint Conveyor	MANUFACTURING	ADVANCED
13	Conveyor Belt to Packing Station	LOGISTICS	EASY
14	Parcel Sorter with Three Chutes	LOGISTICS	MEDIUM
15	Cross-dock: Truck Waves to Outbound Docks	LOGISTICS	ADVANCED
16	Fixed Cable Latency	NETWORK & SIGNAL	EASY
17	One Message, Three Subscribers	NETWORK & SIGNAL	MEDIUM
18	Microservice Gateway Under Load	NETWORK & SIGNAL	ADVANCED

Part B · Pitch catalogue

Five use cases for each of the six domains, tagged Simple or Complex, with the business problem, what to model, the KPIs and the decision it supports.

01 HOW TO USE THIS PLAYBOOK

Read a guide in thirty seconds

Every guide has the same shape: a one-line story, the question it answers, numbered build steps with the exact panel settings, a connection map, the run length, the numbers to expect, and one experiment.

Eight rules before you build

1 · Everything is in seconds
Arrival Speed is per second (72 an hour = 0.02). Times are seconds (4 minutes = 240). Pick the run length with the unit selector next to Duration.

2 · Name every block
Rename each block to its job. Its ID (for example `node_5`) is shown at the top of its settings panel; Decision Blocks need it.

3 · Decision paths need IDs
Draw the arrows, then open the Decision Block → Outgoing Paths → **Add Path**: type the target block's ID and a chance from 0 to 1. Chances must add up to 1, or items stop there.

4 · A Waiting Line in front of every limited block
That is where waits and give-ups are measured. Patience works only when the line feeds a limited block. Line capacity and "who gets served first" are not applied by the engine today; leave them at the defaults.

5 · Time steps vs limited steps
Conveyor, Processing, Assembly and Counter steps hold any number at once: use them for belts, ovens, travel and treatment. Use a limited block when only N can work at the same time.

6 · Fold downtime into the cycle
For a machine available 90% of the time, enter cycle ÷ 0.9. The panel's breakdown fields are not applied by the main engine yet.

7 · Tanks fill, they don't drain
Each arriving batch adds its Fill Rate; a batch that does not fit is dropped. Use tanks for "when does it overflow" questions.

8 · Network guides start from templates
Channel delay and buffer size are set by the template gallery; the settings panel edits sources, decisions and exits.

Three sanity formulas

Utilization

$$\rho = \lambda \times S \div c$$

Arrival rate × time per item ÷ number of servers. ρ ≥ 1 means the line grows forever.

Throughput

$$out \approx \lambda \times T$$

When ρ < 1, everything that arrives leaves, minus what is still inside at the end.

Little's Law

$$L = \lambda \times W$$

Items inside a step = arrival rate × time spent there. Sizes belts, ovens and buffers.

02 GLOSSARY

Same engine, domain-specific names

Each domain's palette uses its own words for the same few building blocks. Learn the eight behaviours once and every domain reads the same.

PALETTE NAME	WHERE	WHAT IT DOES AND WHAT YOU SET
Start Point	All domains	Creates arrivals: **Paced Arrivals** (pattern + Arrival Speed per second) or an **Exact Time Schedule** (at second N, release M). Several outgoing arrows take turns.
Waiting Line	People, Vehicle, Manufacturing, Logistics	Where items wait in front of a limited block; wait time and give-ups (**Will People Give Up**, patience in seconds) are measured here.
Limited Counter · Service Bay (Limited) · Machine / Robot · Loading Dock	One per domain	The same limited resource: **Number of Staff / Parallel Counters**, **Average Time to Serve** (s), **Service Speed Consistency**. Only N items at once.
Counter · Processing Block · Processing Unit · Assembly Step · Conveyor Step	One per domain	A time-only step: **Average Duration** (s) and **Speed Consistency**. Holds any number at once, so use it for belts, ovens, travel and treatment time.
Decision Block · Valve / Splitter	All domains	Sends each item down one path by chance. Set under **Outgoing Paths** → **Add Path**: target block ID and a chance from 0 to 1.
Storage Tank	Liquid	Accumulates **Fill Rate** units per arriving batch up to **Capacity**; batches that do not fit are dropped.
Transport Channel · Storage Rack · Broadcaster	Network (Storage Rack also in Liquid)	Delay link with a buffer · holding buffer · copy to every path. Their settings come from templates.
Exit Point	All domains	Counts completed items; **Record Final Score** includes them in the report.

Reading the numbers

Where the "Measured" numbers come from

Every guide was built on a real JustCmul8 canvas (the screenshots), then run 10 times with the exact SimPy script the app generates for it (the same code the in-browser engine runs), on SimPy 4.1.2. Measured values are the mean of those 10 runs; experiments use 5 runs.

Theory values move

Expected numbers are steady-state theory. A single run varies, more so near 80–90% load. When a figure is far off (for example utilization 60% where 15% is expected), it is worth a bug report; when it is within the normal spread, run more replications in Reliability Lab.

A

Step-by-step guides

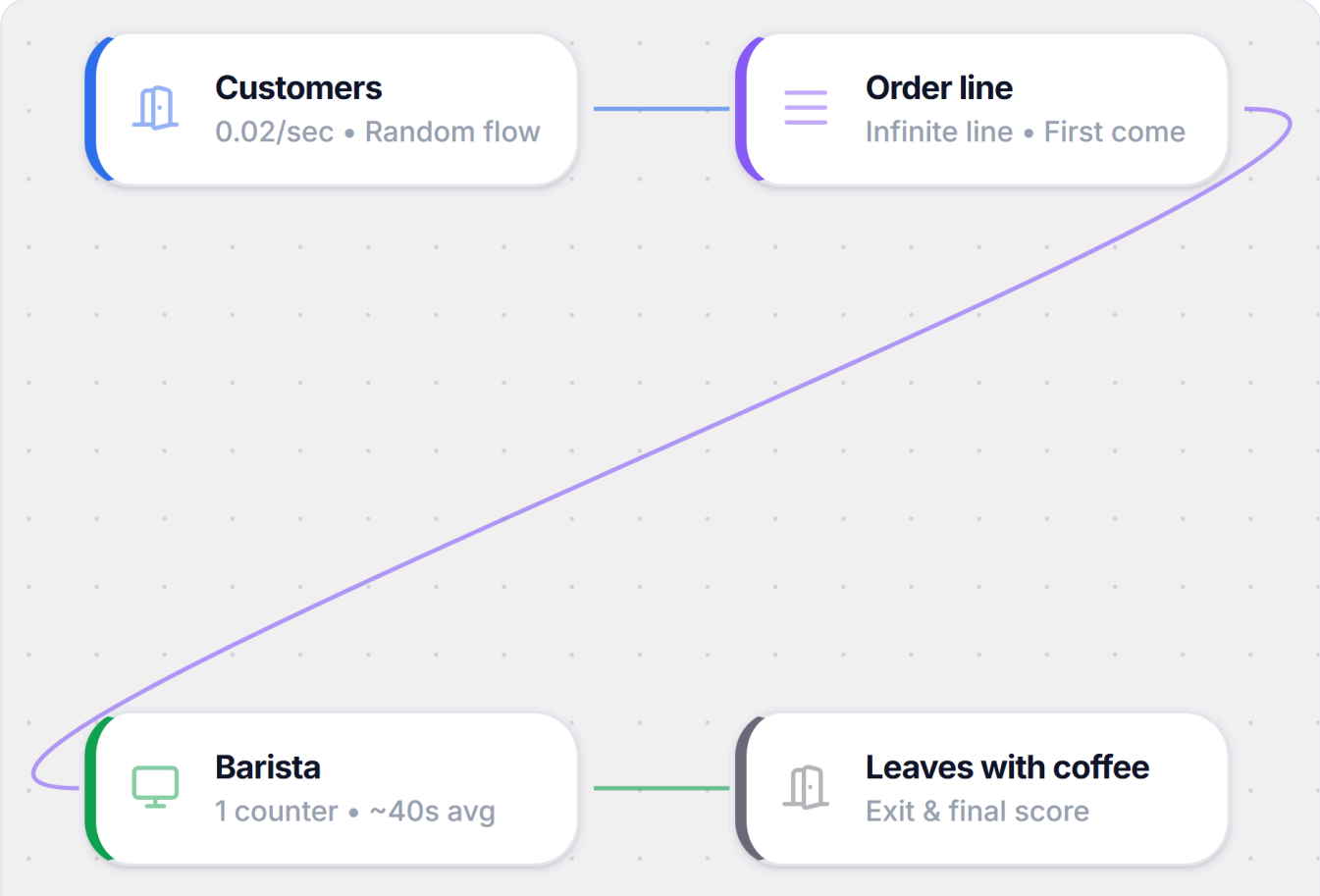
Eighteen builds: Easy, Medium and Advanced in each of the six domains. Each takes two pages: the design as it looks on the canvas, then the build steps and the numbers to check.

GUIDE 01 PEOPLE & SERVICE EASY

Coffee Shop Counter

A café with one barista. In the morning rush a customer walks in about every 50 seconds.

QUESTION Is one barista enough, or does the line get out of hand?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Customers → Order line → Barista → Leaves with coffee
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	4	One server near 80% busy already makes people wait minutes.

GUIDE 01 PEOPLE & SERVICE EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Customers**	Paced Arrivals · Arrival Pattern: **Natural Random Spacing** · Arrival Speed: **0.02**	0.02 per second = 72 customers an hour. Random gaps, because people don't arrive on a timetable.
2	**Waiting Line → Limited Counter**	**Order line → Barista**	Line: defaults · Number of Staff: **1** · Average Time to Serve: **40** · Consistency: **Random Service Duration**	One person, and some orders are quick while others take longer.
3	**Exit Point**	**Leaves with coffee**	Record Final Score: **On**	Customers are counted here.

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Customers arrived	≈ 576	**577** (530–635)	Random arrivals: ±10% run to run.
Barista utilization	≈ 80%	**79%**	Matches.
Average wait in line	≈ 160 s	**157 s**	Matches; single runs swing ±40%.
Served	≈ arrivals	**574**	The last few are still in line or at the counter.

ON THE RESULTS DASHBOARD

- **Executive Summary**: the Bottleneck card names **Order line**.
- **Line Buildup Timeline**: the line rises and falls in bursts; it never settles at a fixed length.
- **Station Counters**: Barista busy ≈ 79%.

COMMON MISTAKES

- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.
- Skipping the Waiting Line. Without it, waits are still counted on the limited block, but patience (give-ups) cannot work.
- Using **Fixed & Exact Spacing** for customers: arrivals become a clock and the line disappears, which is not what a café sees.

EXPERIMENT

Try: Set Number of Staff to **2**.

Result: Theory: 40% busy, wait ≈ 8 s. **Measured: 42%, 9 s.** Doubling staff cut waiting by about 95%, not 50%.

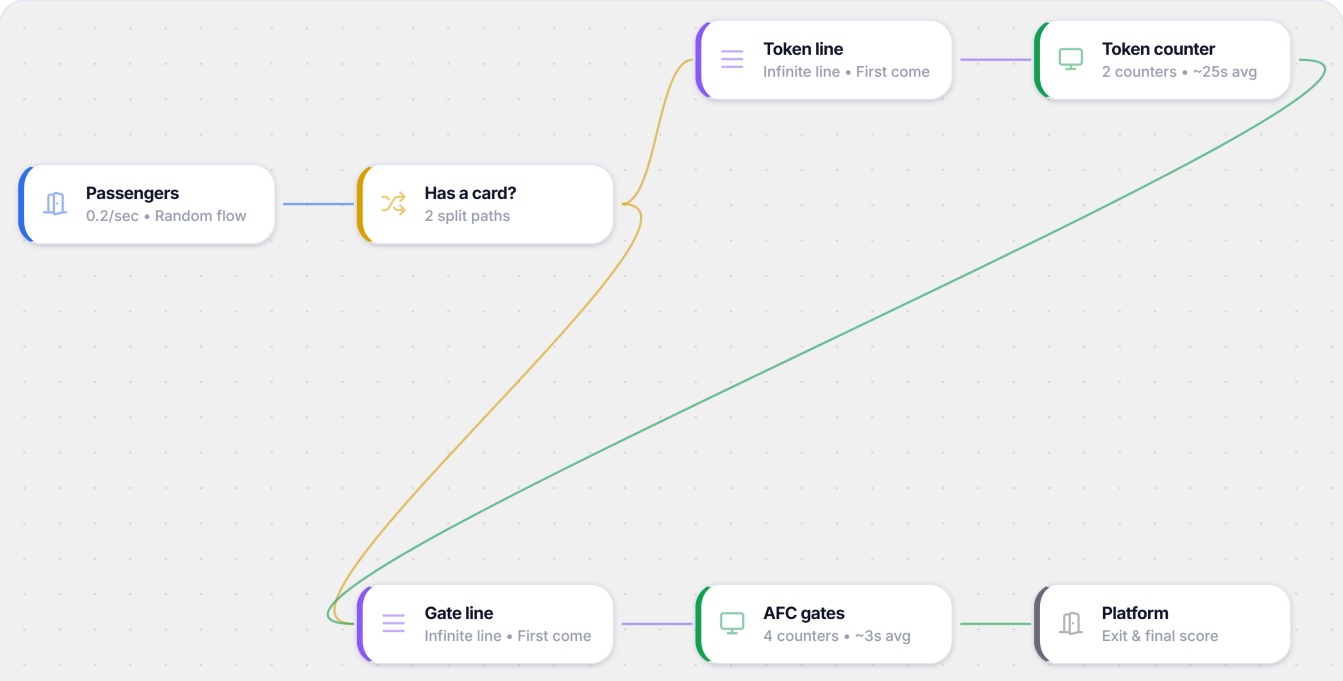
> At 80% load a single run swings a lot (±40% on the wait). Use Reliability Lab → Stress with several runs for a stable average.

GUIDE 02 PEOPLE & SERVICE MEDIUM

Metro Station Entry

Evening peak at a metro station. A passenger arrives about every 5 seconds; 40% have no card and must buy a token first.

QUESTION How many token counters stop the line from growing, and how many passengers give up today?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Passengers → Has a card?
Has a card? → (0.4) Token line → Token counter → Gate line
Has a card? → (0.6) Gate line → AFC gates → Platform
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	7	Exactly 100% load is still overload: the line has no steady length.

GUIDE 02 PEOPLE & SERVICE MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Passengers**	Arrival Pattern: **Natural Random Spacing** · Arrival Speed: **0.2**	One every 5 seconds, about 720 an hour.
2	**Decision Block**	**Has a card?**	After connecting: Outgoing Paths → Add Path → **Token line 0.4**, **Gate line 0.6**	Each passenger takes one route, by chance. Chances must add up to 1.
3	**Waiting Line**	**Token line**	Will People Give Up: **Random Patience Limits** · Average Patience Time: **300**	Real people leave after a few minutes; this measures how many are lost.
4	**Limited Counter**	**Token counter**	Number of Staff: **2** · Time to Serve: **25** · **Random Service Duration**	A clerk's time varies with each passenger.
5	**Waiting Line → Limited Counter**	**Gate line → AFC gates**	Line: defaults · Number of Staff: **4** · Time to Serve: **3** · **Fixed Constant Duration**	Four gates; a card tap takes the same time every time.
6	**Exit Point**	**Platform**	Record Final Score: **On**	Counts passengers who made it through.

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Passengers arrived	≈ 720	**713** (689–751)	Matches.
Token counter utilization	100% (overloaded)	**85%**	The station starts empty; in one hour the line has not yet grown to its full length, and about 11% give up, which frees the clerks.
Gave up in the token line	> 0, growing	**31** (11% of token buyers)	Patience of 5 minutes at work.
AFC gate utilization	≤ 15%	**14%**	Matches.
Reached the platform	arrivals − give-ups − still queued	**677**	713 − 31 − a few still inside.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Token line**.
- **Line Buildup Timeline**: the token line keeps climbing through the hour.
- **Flow & Balance Check**: arrivals = platform + give-ups + still inside.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Connecting the Token counter straight to the Platform. Token buyers must still pass the gates.
- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.

EXPERIMENT

Try: Set Token counter staff to **3**; compare both versions in Reliability Lab → Compare.

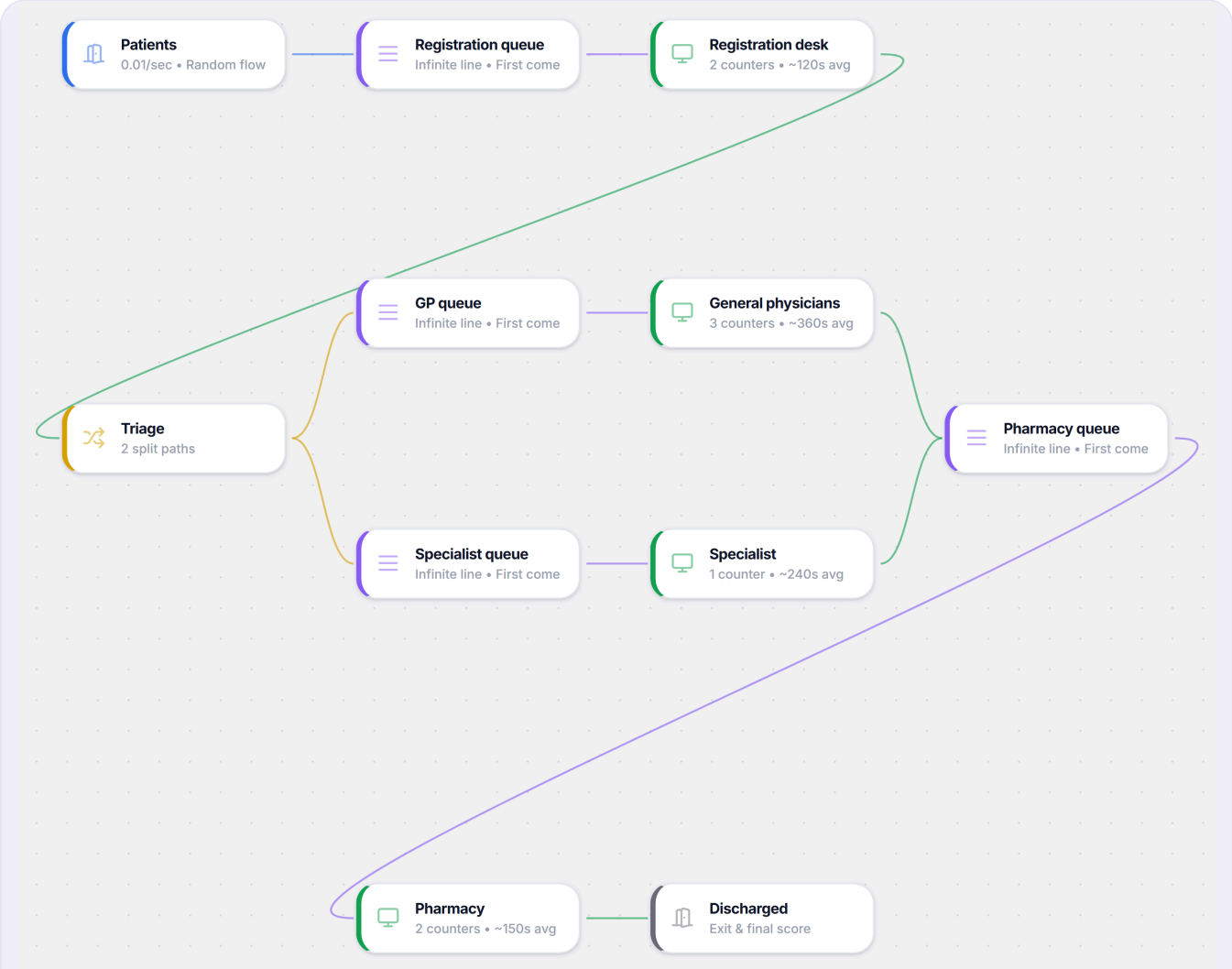
Result: Theory: 67%, wait ≈ 11 s. **Measured: 67%, 8 s, only 7 give up.**

GUIDE 03 PEOPLE & SERVICE ADVANCED

Hospital OPD Patient Flow

An outpatient department sees about 36 patients an hour: registration, then a general physician (70%) or a specialist (30%), then the pharmacy.

QUESTION Where is the bottleneck, and how many general physicians keep the wait under two minutes?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Patients → Registration queue → Registration desk → Triage
Triage → (0.7) GP queue → General physicians → Pharmacy queue
Triage → (0.3) Specialist queue → Specialist → Pharmacy queue
Pharmacy queue → Pharmacy → Discharged
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	11	Patience caps waits: the people who leave are the cost.

GUIDE 03 PEOPLE & SERVICE ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Patients**	**Natural Random Spacing** · Arrival Speed: **0.01**	36 an hour.
2	**Waiting Line → Limited Counter**	**Registration queue → Registration desk**	Line: defaults · Staff **2** · Time **120** · **Random Service Duration**	Two clerks, two minutes each on average.
3	**Decision Block**	**Triage**	Paths: **GP queue 0.7**, **Specialist queue 0.3**	Triage sends each patient one way.
4	**Waiting Line**	**GP queue**	**Random Patience Limits** · Patience **1800**	Some patients leave after about 30 minutes.
5	**Limited Counter**	**General physicians**	Staff **3** · Time **360** · **Random Service Duration**	Six-minute consultations.
6	**Waiting Line → Limited Counter**	**Specialist queue → Specialist**	Line: defaults · Staff **1** · Time **240** · **Random Service Duration**	
7	**Waiting Line → Limited Counter**	**Pharmacy queue → Pharmacy**	Line: defaults · Staff **2** · Time **150** · **Random Service Duration**	Both doctor paths merge here.
8	**Exit Point**	**Discharged**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Patients arrived	≈ 288	**292** (265–325)	Matches.
Registration	60% · 68 s	**61% · 72 s**	Matches.
General physicians	84% · 535 s (no give-ups)	**75% · 167 s · 19 leave**	Theory ignores patience. With a 30-minute patience about 10% leave, which trims both the load and the longest waits.
Specialist	72% · 617 s	**69% · 493 s**	Only ≈ 86 patients in 8 hours starting empty: a short, noisy sample.
Pharmacy	75% · 193 s	**67% · 141 s**	Fewer patients reach it because some left the GP queue.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Specialist queue**, the longest average wait.
- **Station Counters**: GP queue shows the give-up count.
- **Person / Item Journey**: follow one patient through both branches.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Patience on the wrong line. Put it on the GP queue that feeds the physicians; on a line that feeds a Decision Block it does nothing.
- Forgetting that both doctor branches must connect to the same Pharmacy queue.

EXPERIMENT

Try: Compare **3 vs 4** general physicians in Reliability Lab → Compare.

Result: Theory: 63%, 79 s. **Measured: 60%, 32 s, 5 leave.** The Specialist queue (≈ 460 s) is now clearly the next decision.

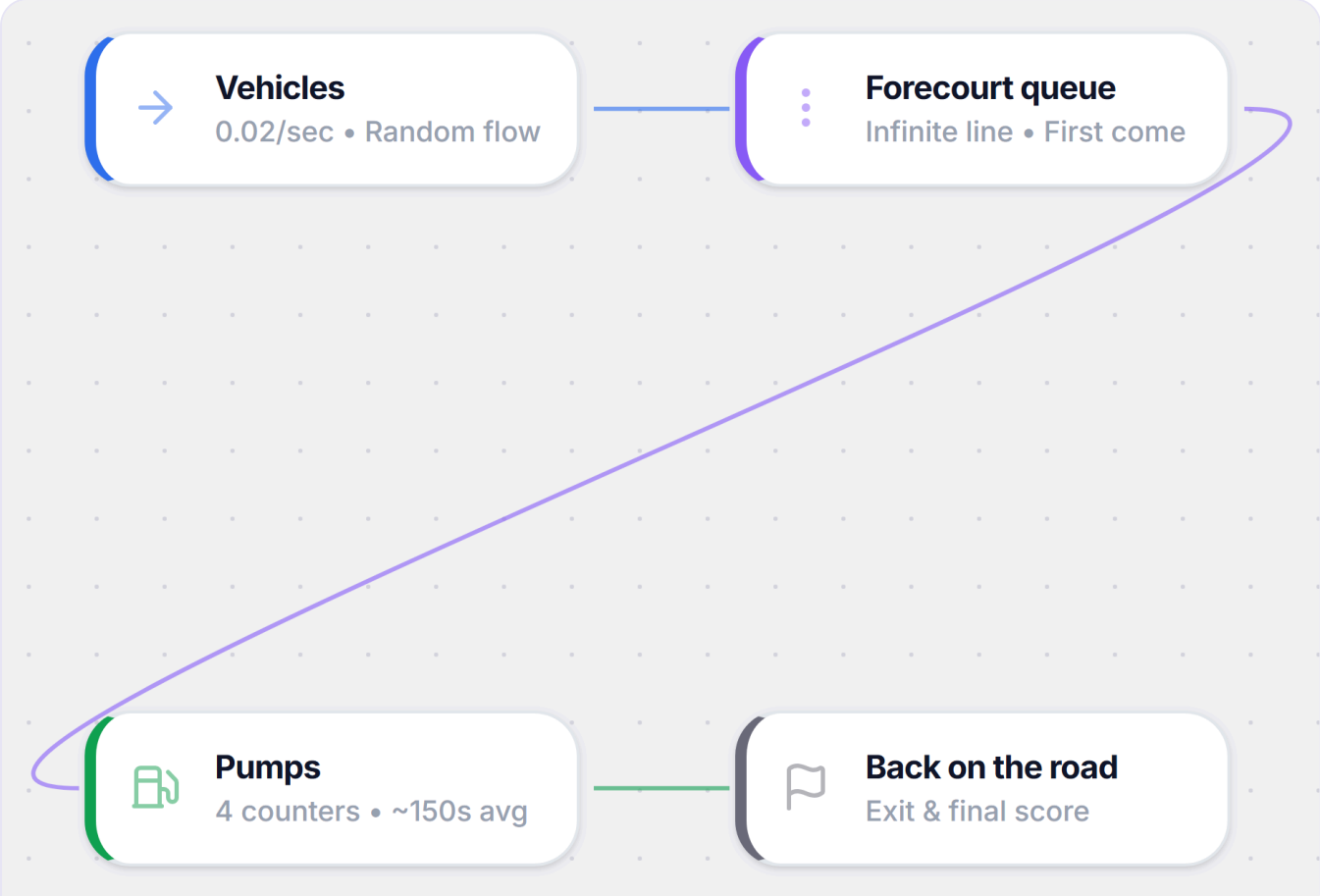
> The clinic starts empty, so the first half hour reads light; an 8-hour run lands slightly below these steady-state figures.

GUIDE 04 TRAFFIC & VEHICLES EASY

Fuel Station Pumps

A highway fuel station sees about 72 vehicles an hour; a fill-up takes 2½ minutes on average.

QUESTION How many pumps does the station need?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

`Vehicles → Forecourt queue → Pumps → Back on the road`

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	4	Pump count is set by the busiest hour, not the daily average.

GUIDE 04 TRAFFIC & VEHICLES EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Vehicles**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 an hour.
2	**Waiting Line → Service Bay (Limited)**	**Forecourt queue → Pumps**	Line: defaults · Staff/Parallel: **4** · Time **150** · **Random Service Duration**	Four pumps, each serving one vehicle at a time.
3	**Exit Point**	**Back on the road**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Vehicles	≈ 576	**578** (535–623)	Matches.
Pump utilization	≈ 75%	**74%**	Matches.
Average wait	≈ 76 s	**71 s**	Matches.
Back on the road	≈ arrivals	**573**	A few still at the pumps.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Forecourt queue**.
- **Station Counters**: Pump/Bay Util. ≈ 74%.
- **Cost & ROI**: add an hourly cost to the pumps to price the decision.

COMMON MISTAKES

- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.
- Using four separate one-pump bays instead of one block with 4 parallel pumps; one shared queue is how a forecourt behaves.
- Skipping the Waiting Line. Without it, waits are still counted on the limited block, but patience (give-ups) cannot work.

EXPERIMENT

Try: Drop to **3** pumps and set Exact Maximum Wait Limit **300** on the queue.

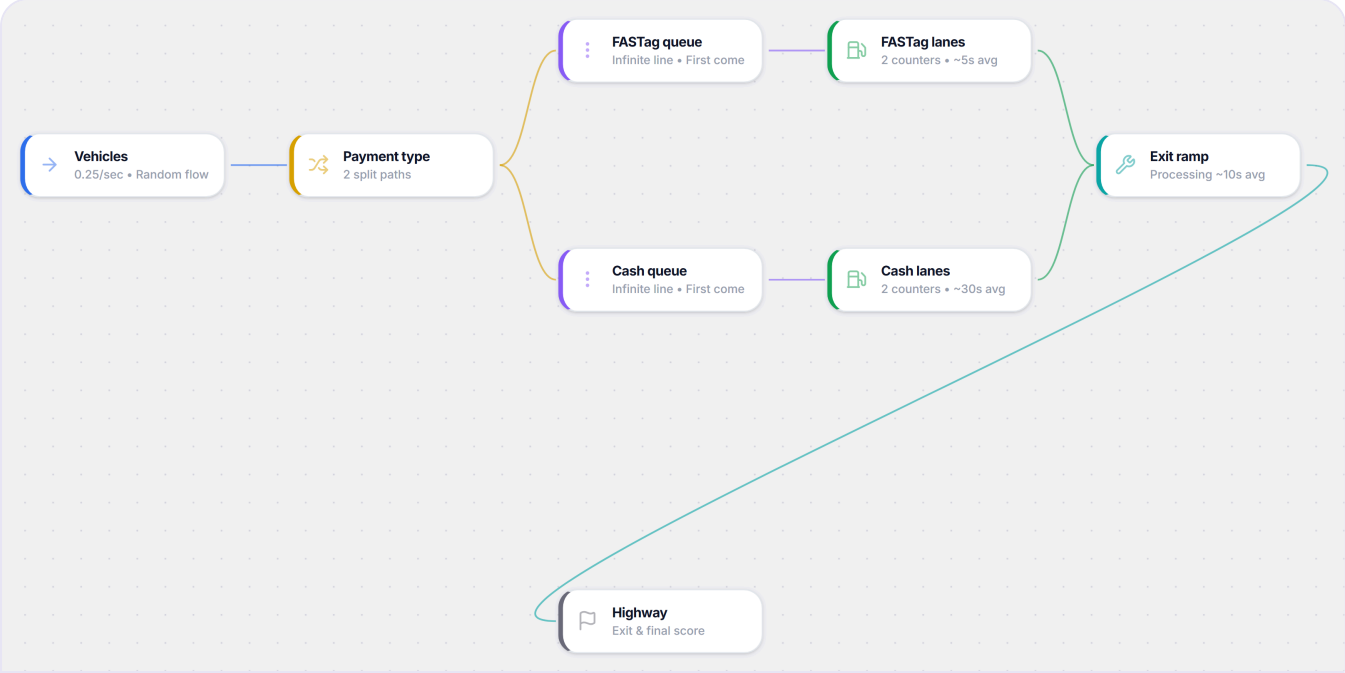
Result: Theory: 100% load, the queue never settles. **Measured: 87% busy, 110 s wait, and 62 drivers (11%) leave.** Those are lost sales.

GUIDE 05 TRAFFIC & VEHICLES MEDIUM

Toll Plaza: FASTag vs Cash

A toll plaza handles about 900 vehicles an hour; 80% pay by FASTag, 20% in cash.

QUESTION How many cash lanes are needed, and what happens if FASTag use drops?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Vehicles → Payment type
Payment type → (0.8) FASTag queue → FASTag lanes → Exit ramp
Payment type → (0.2) Cash queue → Cash lanes → Exit ramp
Exit ramp → Highway
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	8	A policy shift (tag adoption) can overload infrastructure overnight.

GUIDE 05 TRAFFIC & VEHICLES MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Vehicles**	**Natural Random Spacing** · Arrival Speed: **0.25**	900 an hour.
2	**Decision Block**	**Payment type**	Paths: **FASTag queue 0.8**, **Cash queue 0.2**	
3	**Waiting Line → Service Bay (Limited)**	**FASTag queue → FASTag lanes**	Line: defaults · **2** · Time **5** · **Fixed Constant Duration**	A tag read takes the same time every time.
4	**Waiting Line → Service Bay (Limited)**	**Cash queue → Cash lanes**	Line: defaults · **2** · Time **30** · **Random Service Duration**	Cash handling varies.
5	**Processing Block**	**Exit ramp**	Average Duration **10** · **Fixed Constant Duration**	Travel time only; any number of cars can be on the ramp.
6	**Exit Point**	**Highway**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Vehicles	≈ 900	**893** (839–949)	Matches.
FASTag lanes	≈ 50%	**49%**	Matches.
Cash lanes	75% · 39 s	**74% · 30 s**	One-hour runs start empty, so waits read a little low.
Exit ramp time	10 s	**10.0 s**	A time-only step: every car takes exactly 10 s.
Reached the highway	≈ arrivals	**887**	A few still on the ramp.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Cash queue**.
- **Traffic Heatmap**: the cash side runs hot, the FASTag side cool.
- **Line Buildup Timeline**: compare both queues.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Using a Service Bay for the exit ramp. The ramp is travel time with no limit; a limited bay would invent a queue.
- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.

EXPERIMENT

Try: Change the split to **0.6 / 0.4**, then try **4** cash lanes.

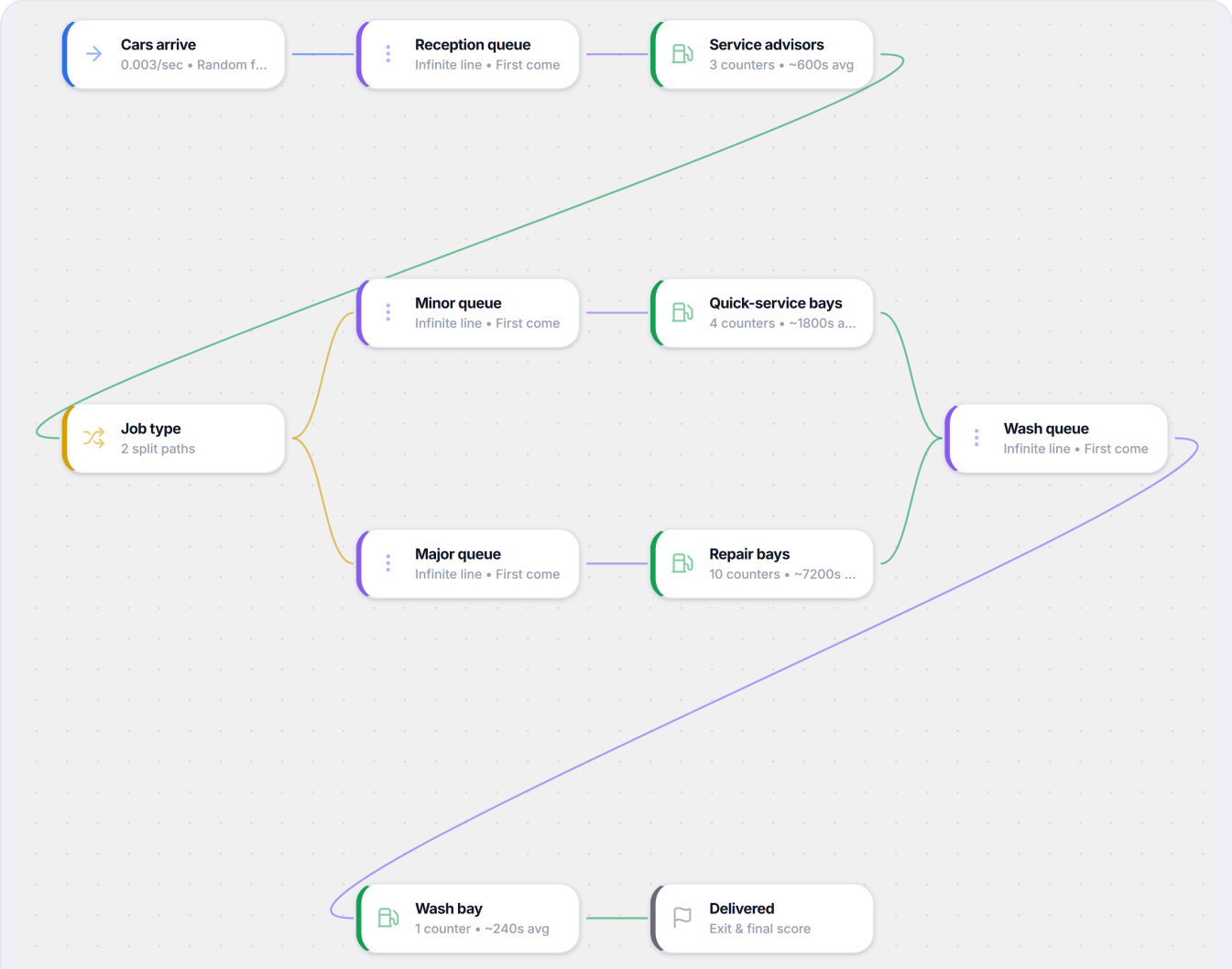
Result: Theory: 150% load, then 75% and 15 s. **Measured: 99% busy with a 635 s average wait and climbing; with 4 lanes 79% and 18 s.**

GUIDE 06 TRAFFIC & VEHICLES ADVANCED

Car Service Centre

A workshop receives about 11 cars an hour: advisors take the job, 60% get a quick service and 40% a major repair, then every car is washed.

QUESTION Which bays limit the workshop, and how much demand growth can it absorb?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Cars arrive → Reception queue → Service advisors → Job type
Job type → (0.6) Minor queue → Quick-service bays → Wash queue
Job type → (0.4) Major queue → Repair bays → Wash queue
Wash queue → Wash bay → Delivered
```

RUN LENGTH	BLOCKS	KEY IDEA
3 days (two-hour repairs make a one-day run mostly warm-up)	11	Long jobs need long runs: two-hour repairs take days to settle.

GUIDE 06 TRAFFIC & VEHICLES ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Cars arrive**	**Natural Random Spacing** · Arrival Speed: **0.003**	About 10.8 cars an hour.
2	**Waiting Line → Service Bay (Limited)**	**Reception queue → Service advisors**	Line: defaults · **3** · Time **600** · **Random Service Duration**	Ten minutes to write up a job.
3	**Decision Block**	**Job type**	Paths: **Minor queue 0.6**, **Major queue 0.4**	
4	**Waiting Line → Service Bay (Limited)**	**Minor queue → Quick-service bays**	Line: defaults · **4** · Time **1800** · **Random Service Duration**	30-minute services.
5	**Waiting Line → Service Bay (Limited)**	**Major queue → Repair bays**	Line: defaults · **10** · Time **7200** · **Random Service Duration**	Two-hour repairs.
6	**Waiting Line → Service Bay (Limited)**	**Wash queue → Wash bay**	Line: defaults · **1** · Time **240** · **Random Service Duration**	Both paths merge.
7	**Exit Point**	**Delivered**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Cars	≈ 778	**769** (707–834)	Matches.
Service advisors	60% · 177 s	**59% · 149 s**	Matches within run-to-run spread.
Quick-service bays	81% · 24 min	**78% · 20 min**	Matches within spread.
Repair bays	86% · 50 min	**81% · 23 min**	Two-hour jobs need a long run. A 30-day run measures 85% · 37 min, converging on theory.
Wash bay	72% · 10 min	**70% · 11 min**	Matches.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Major queue**.
- **Station Counters**: compare the two bay pools side by side.
- **Line Buildup Timeline**: the repair queue builds slowly over days, which is why a 3-day run is the minimum.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Running for 8 hours: with two-hour repairs the bays barely fill and every figure reads low.
- Splitting a pool into separate one-bay blocks; use one block with the number of bays.

EXPERIMENT

Try: Reliability Lab → Stress: demand **+15%** (Arrival Speed 0.00345), then try **11** repair bays.

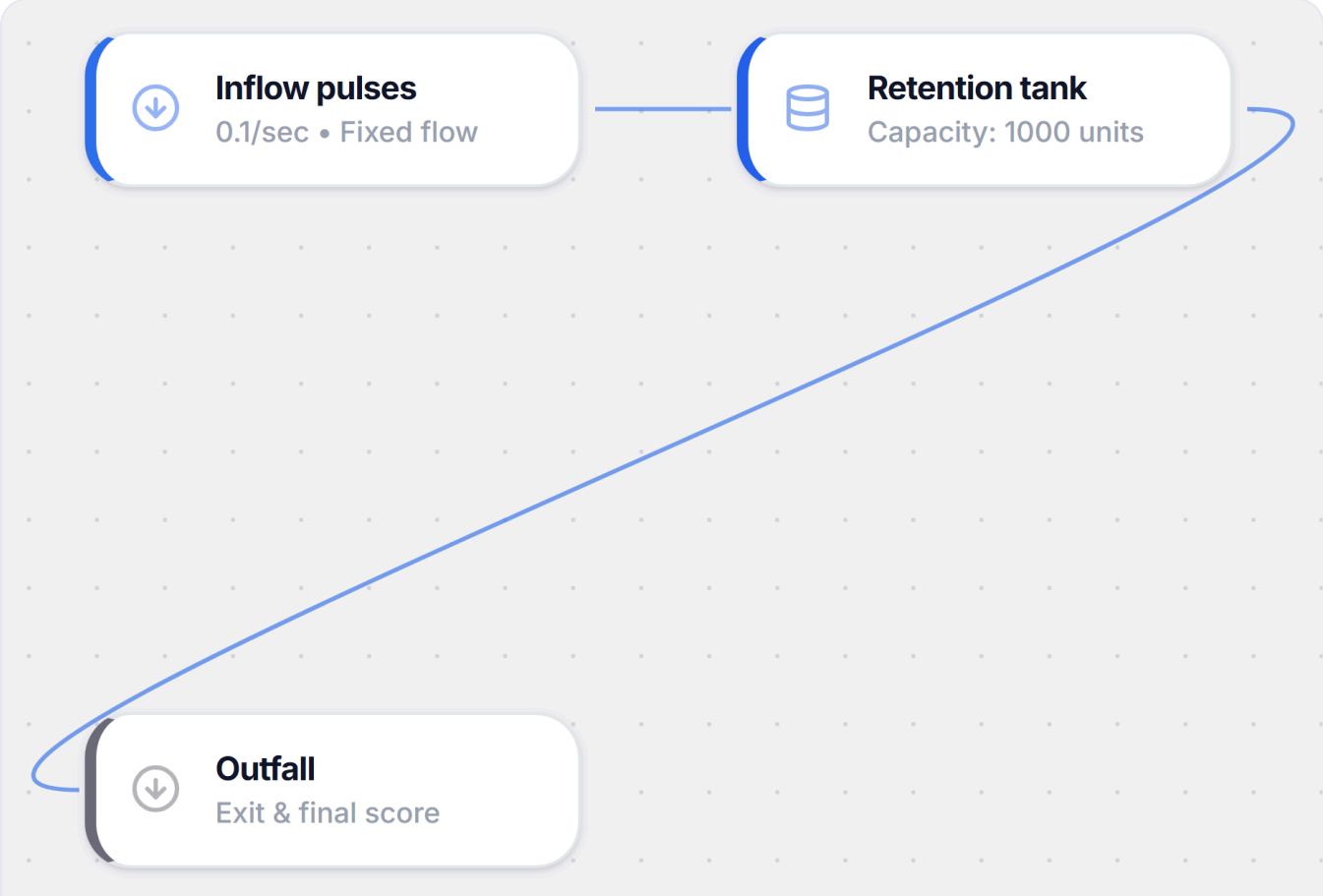
Result: Measured: repair bays 94% and a 2.5-hour wait, quick-service bays 92%. An 11th repair bay brings repairs to 88%, but quick service stays at 91%: growth needs both pools.

GUIDE 07 LIQUID & MATERIAL EASY

Storm-water Retention Tank

During a storm, inflow pulses reach a retention tank every 10 seconds, each adding 20 units.

QUESTION How long until the tank overflows if nothing is drawn off?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Inflow pulses → Retention tank → Outfall
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	3	A tank with no outflow is a timer: capacity ÷ inflow.

GUIDE 07 LIQUID & MATERIAL EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inflow pulses**	**Fixed & Exact Spacing** · Arrival Speed: **0.1**	Exactly one pulse every 10 seconds.
2	**Storage Tank**	**Retention tank**	Capacity **1000** · Initial Level **0** · Fill Rate **20**	In the engine, each arriving pulse adds the Fill Rate amount to the tank.
3	**Exit Point**	**Outfall**	Record Final Score: **On**	Counts pulses the tank accepted.

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Pulses arriving	359	**359**	The run stops at exactly 3,600 s, so the pulse due at 3,600 s is not counted.
Tank full after	50 pulses (500 s)	**50 accepted · level 1,000**	Exact: everything here is fixed.
Pulses dropped	309	**309**	Exact.
Tank utilization	100%	**100%**	Tank utilization is level ÷ capacity.

ON THE RESULTS DASHBOARD

- **Station Counters**: Retention tank shows level 1,000 and 309 dropped.
- **Line Buildup Timeline**: the tank level climbs in steps and flattens at 500 s.
- **Flow & Balance Check**: in 359 = out 50 + dropped 309.

COMMON MISTAKES

- Expecting the tank to drain. In the current engine tanks only fill; model outflow as a separate path.
- Reading Fill Rate as units per second. The engine adds it once per arriving pulse.
- Leaving Capacity blank: the panel shows 100 but the engine uses 1,000. Always type it.

EXPERIMENT

Try: Set Capacity to **5000**.

Result: Theory: 250 pulses fit (≈ 42 min). **Measured: 250 accepted, 109 dropped.**

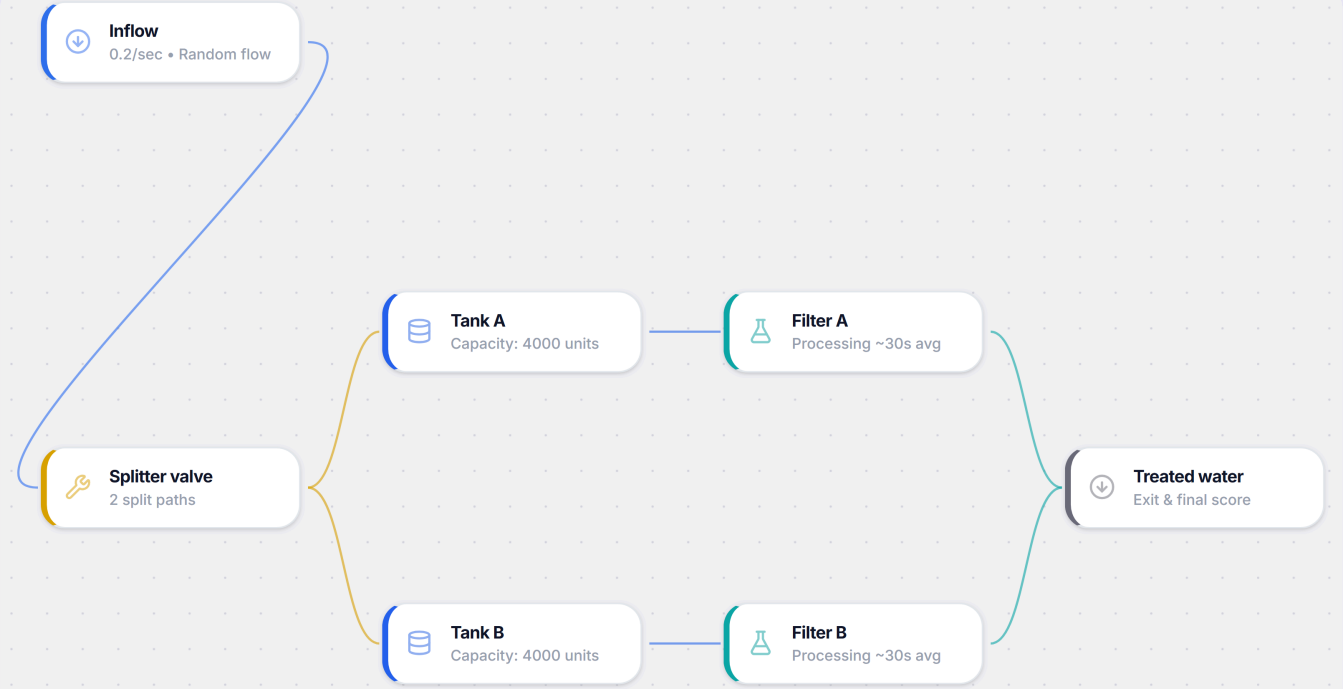
> Tanks fill but are not drained in the current engine; a pulse that does not fit is counted as dropped and never reaches the exit.

GUIDE 08 LIQUID & MATERIAL MEDIUM

Two Tanks, One Valve

A splitter valve sends 70% of incoming batches to Tank A and 30% to Tank B; each tank feeds a filter.

QUESTION Which tank overflows first, and what split avoids overflow for an hour?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Inflow → Splitter valve
Splitter valve → (0.7) Tank A → Filter A → Treated water
Splitter valve → (0.3) Tank B → Filter B → Treated water
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	7	Unequal splits overflow the busy side first.

GUIDE 08 LIQUID & MATERIAL MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inflow**	**Natural Random Spacing** · Arrival Speed: **0.2**	720 batches an hour.
2	**Valve / Splitter**	**Splitter valve**	Paths: **Tank A 0.7**, **Tank B 0.3**	
3	**Storage Tank**	**Tank A**	Capacity **4000** · Initial **0** · Fill Rate **10**	
4	**Storage Tank**	**Tank B**	Capacity **4000** · Initial **0** · Fill Rate **10**	
5	**Processing Unit**	**Filter A**	Average Duration **30** · **Random Service Duration**	Treatment time; many batches can be in the filter at once.
6	**Processing Unit**	**Filter B**	Average Duration **30** · **Random Service Duration**	
7	**Exit Point**	**Treated water**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Batches	≈ 720	**724** (701–743)	Matches.
Tank A	full ≈ 48 min · ≈ 104 dropped	**full · 114 dropped**	Matches.
Tank B	≈ 54%	**53%**	Matches.
Treated water	≈ 616	**609**	Matches.

ON THE RESULTS DASHBOARD

- **Station Counters**: Tank A shows dropped batches, Tank B none.
- **Line Buildup Timeline**: Tank A's level flattens around minute 48.
- **Flow & Balance Check**: dropped batches explain the gap between in and out.

COMMON MISTAKES

- Valve / Splitter: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Expecting the Filters to limit flow. Processing Units have no capacity limit; they only add time.
- Leaving tank capacity blank (see Guide 07).

EXPERIMENT

Try: Change the split to **0.5 / 0.5**.

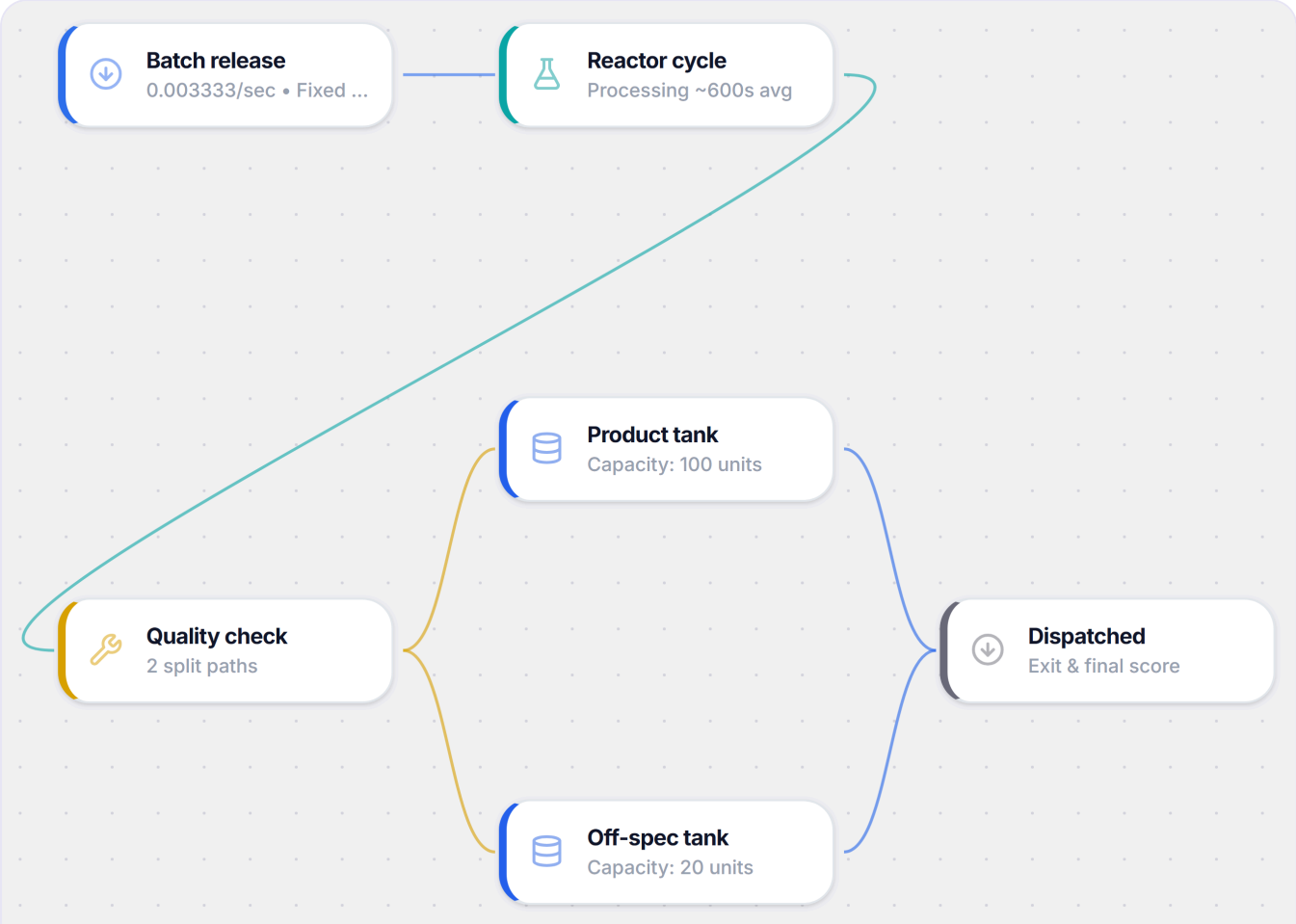
Result: Theory: ≈ 90% each, nothing dropped. **Measured: 88% and 91%, 0 dropped, 707 treated.**

GUIDE 09 LIQUID & MATERIAL ADVANCED

Batch Reactor with Quality Split

A plant releases a batch every 5 minutes into a 10-minute reactor cycle; 90% pass QC to the product tank, 10% go to an off-spec tank.

QUESTION How many reactors does the schedule need, and when do the tanks fill?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Batch release → Reactor cycle → Quality check
Quality check → (0.9) Product tank → Dispatched
Quality check → (0.1) Off-spec tank → Dispatched
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	6	Little's Law sizes equipment: vessels = release rate × cycle time.

GUIDE 09 LIQUID & MATERIAL ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Batch release**	**Fixed & Exact Spacing** · Arrival Speed: **0.003333**	One batch every 300 seconds.
2	**Processing Unit**	**Reactor cycle**	Average Duration **600** · **Fixed Constant Duration**	A Processing Unit has no limit on simultaneous batches, so its average contents tell you how many vessels you need.
3	**Valve / Splitter**	**Quality check**	Paths: **Product tank 0.9**, **Off-spec tank 0.1**	
4	**Storage Tank**	**Product tank**	Capacity **100** · Initial **0** · Fill Rate **1**	Fill Rate 1 counts batches.
5	**Storage Tank**	**Off-spec tank**	Capacity **20** · Initial **0** · Fill Rate **1**	
6	**Exit Point**	**Dispatched**	Record Final Score: **On**	Connect both tanks to it.

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Batches released	≈ 95–96	**95**	0.003333 per second is one every 300.03 s.
Batches in the reactor at once	2	**1.97**	Little's Law: (1 ÷ 300) × 600 = 2 vessels. Measured as the time-averaged count in the reactor.
Completed	≈ 93–94	**93**	The last two are still cooking at the end.
Product tank	≈ 84 (84%)	**84 (84%)**	Matches.
Off-spec tank	≈ 9 (47%)	**9 (46%)**	Matches.

ON THE RESULTS DASHBOARD

- **Station Counters**: Reactor cycle in/out tells you how many are inside.
- **Line Buildup Timeline**: product tank level rises in a straight line.
- **Person / Item Journey**: one batch shows the 600 s cycle exactly.

COMMON MISTAKES

- Reading the Processing Unit's utilization: with unlimited capacity it reads 100% and means nothing. Use Little's Law.
- Valve / Splitter: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Release rate typed as minutes (5) instead of per second (0.003333).

EXPERIMENT

Try: Release every 2 minutes: Arrival Speed **0.008333**.

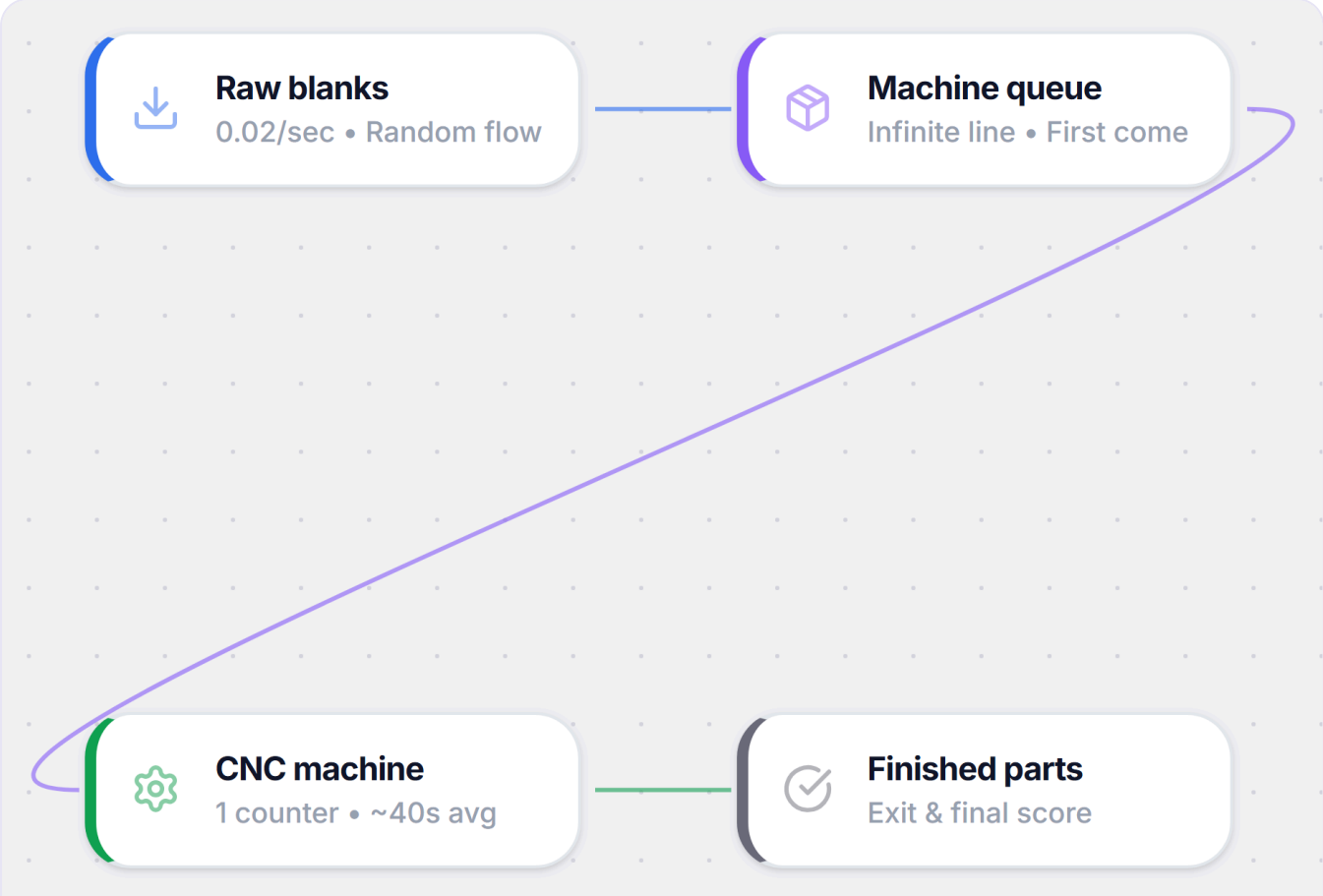
Result: Theory: 5 reactors; product tank overflows ≈ 3.9 h. **Measured: product tank full with 112 batches dropped; off-spec 89% full, 4 dropped.**

GUIDE 10 MANUFACTURING EASY

Single CNC Machine

One CNC machine receives a blank about every 50 seconds; each part takes 40 seconds.

QUESTION How busy is the machine, and how long do parts wait?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Raw blanks → Machine queue → CNC machine → Finished parts
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	4	A small slowdown near full load multiplies waiting.

GUIDE 10 MANUFACTURING EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Raw blanks**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 blanks an hour.
2	**Waiting Line → Machine / Robot**	**Machine queue → CNC machine**	Line: defaults · Staff/Parallel **1** · Time **40** · **Consistent Around Average**	Machined cycle times vary only a little (the engine uses ±30% spread).
3	**Exit Point**	**Finished parts**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Blanks	≈ 576	**574** (515–642)	Matches.
Machine utilization	≈ 80%	**79%**	Matches.
Average wait	≈ 87 s	**85 s**	Matches.
Cycle time	40 s	**39.8 s**	Consistent Around Average uses a ±30% spread.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Machine queue**.
- **Station Counters**: Machine Utilization ≈ 79%.
- **Line Buildup Timeline**: WIP Count shows the buffer breathing.

COMMON MISTAKES

- Skipping the Waiting Line. Without it, waits are still counted on the limited block, but patience (give-ups) cannot work.
- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.
- Choosing Random Service Duration for a machine: machined cycles are steady; random durations double the waits.

EXPERIMENT

Try: A worn tool: cycle **45** s.

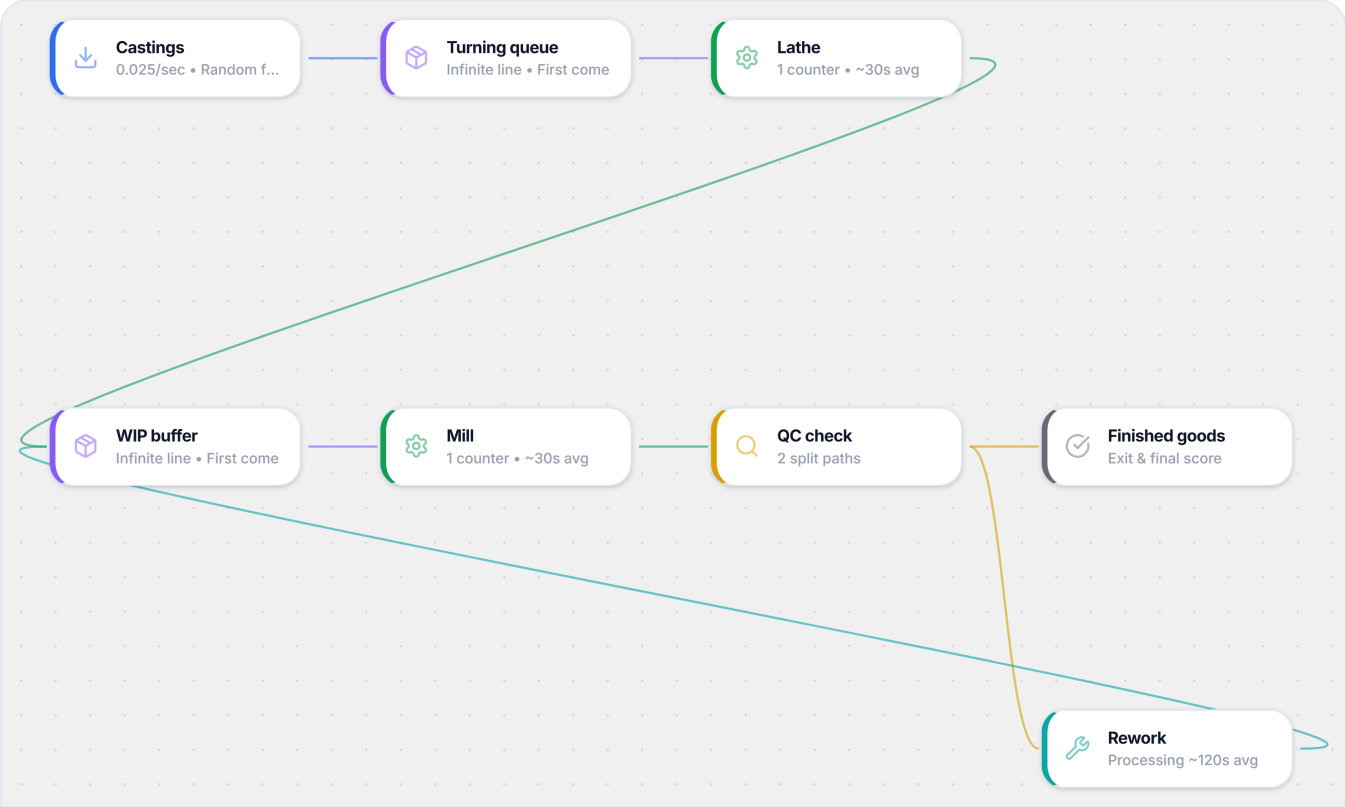
Result: Theory: 90%, 221 s. **Measured: 91%, 332 s**. At 90% load an 8-hour run swings widely; the point holds: 12.5% slower, about 3× the wait.

GUIDE 11 MANUFACTURING MEDIUM

Two-Machine Line with QC Rework

Castings are turned on a lathe, buffered, milled, then checked; 10% fail QC and are reworked before going back to the mill.

QUESTION How much mill capacity does rework consume?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Castings → Turning queue → Lathe → WIP buffer → Mill → QC check
QC check → (0.9) Finished goods
QC check → (0.1) Rework → WIP buffer
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	8	Rework is hidden demand on the machine that repeats the work.

GUIDE 11 MANUFACTURING MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Castings**	**Natural Random Spacing** · Arrival Speed: **0.025**	90 an hour.
2	**Waiting Line → Machine / Robot**	**Turning queue → Lathe**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	
3	**Waiting Line → Machine / Robot**	**WIP buffer → Mill**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	Receives from the lathe and from rework.
4	**Decision Block**	**QC check**	Paths: **Finished goods 0.9**, **Rework 0.1**	
5	**Assembly Step**	**Rework**	Average Duration **120** · **Random Service Duration**	Bench rework; several parts at once.
6	**Exit Point**	**Finished goods**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Castings	≈ 720	**713** (670–749)	Matches.
Lathe	75% · 49 s	**74% · 48 s**	Matches.
Mill (with rework)	83% · 82 s	**81% · 66 s**	Matches within spread.
Rework passes	≈ 80	**80**	Matches.
Finished goods	≈ arrivals	**708**	Matches.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **WIP buffer**, the line in front of the mill.
- **Station Counters**: Mill handles ≈ 790 parts for 713 castings: the rework load.
- **Person / Item Journey**: pick a part that looped through Rework.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Sending Rework back to the Mill directly instead of to the WIP buffer; parts then skip the queue statistics.
- Using a Machine for Rework; bench rework is time-only here.

EXPERIMENT

Try: QC pass rate **0.8** (paths 0.8 / 0.2).

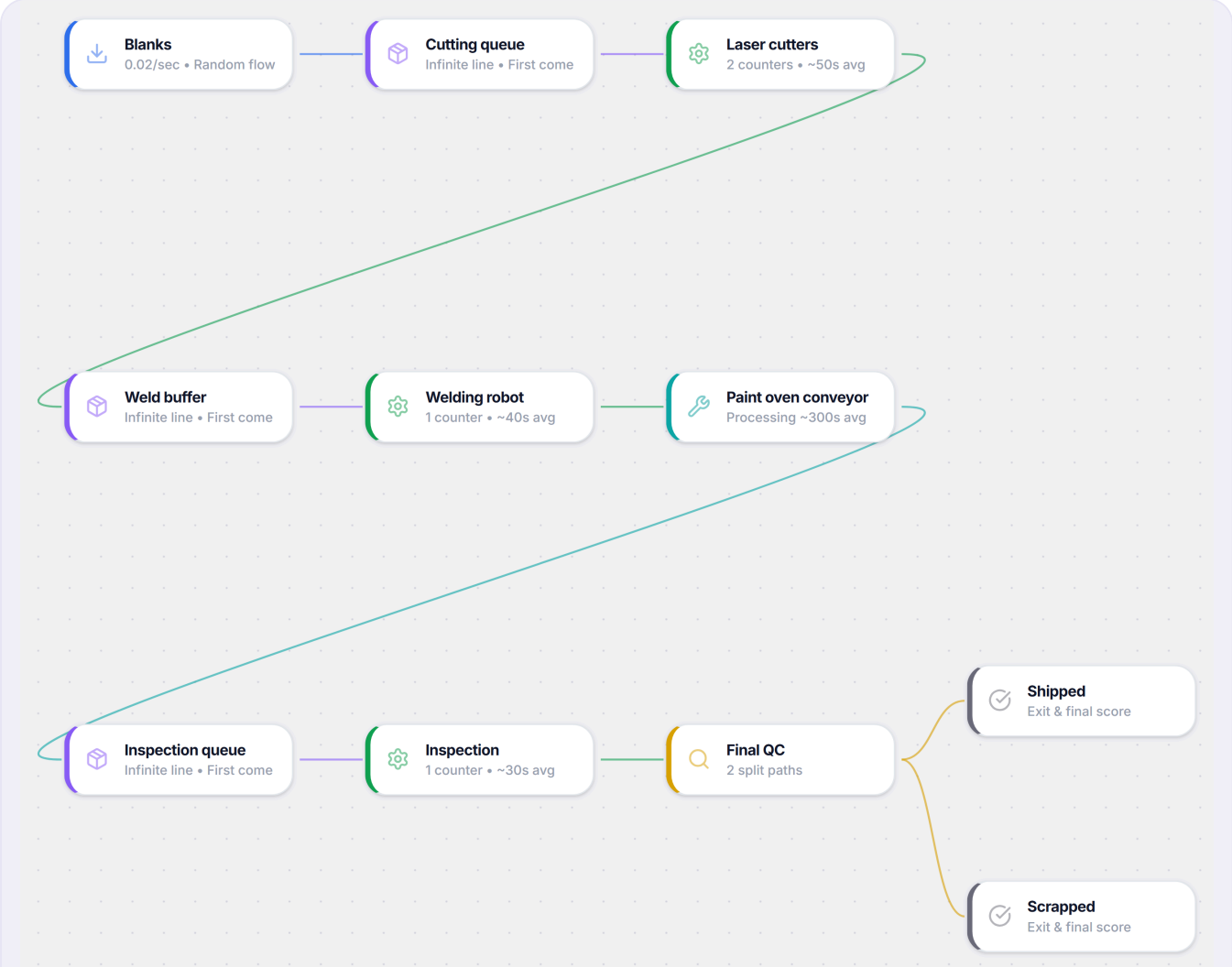
Result: Theory: 94%, 245 s. **Measured: 90%, 140 s**, roughly double the wait. Poor quality eats capacity before it shows as scrap.

GUIDE 12 MANUFACTURING ADVANCED

Welding Bottleneck with Downtime and a Paint Conveyor

Blanks are laser-cut, robot-welded, carried through a paint-curing conveyor oven, inspected, and 5% are scrapped. The welding robot is available 90% of the time.

QUESTION What limits output, and is a second robot or better uptime the better investment?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Blanks → Cutting queue → Laser cutters → Weld buffer → Welding robot
Welding robot → Paint oven conveyor → Inspection queue → Inspection → Final QC
Final QC → (0.95) Shipped · (0.05) Scrapped
```

RUN LENGTH	BLOCKS	KEY IDEA
8 hrs	11	Upstream machines smooth the flow; the bottleneck sets the ceiling.

GUIDE 12 MANUFACTURING ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Blanks**	**Natural Random Spacing** · Arrival Speed: **0.02**	72 an hour.
2	**Waiting Line → Machine / Robot**	**Cutting queue → Laser cutters**	Line: defaults · **2** · Time **50** · **Consistent Around Average**	
3	**Waiting Line → Machine / Robot**	**Weld buffer → Welding robot**	Line: defaults · **1** · Time **40** · **Consistent Around Average**	36 s cycle ÷ 0.90 availability = 40 s. Downtime is folded into the cycle.
4	**Assembly Step**	**Paint oven conveyor**	Average Duration **300** · **Fixed Constant Duration**	A conveyor oven: every part spends 5 minutes inside, many at once.
5	**Waiting Line → Machine / Robot**	**Inspection queue → Inspection**	Line: defaults · **1** · Time **30** · **Consistent Around Average**	
6	**Decision Block**	**Final QC**	Paths: **Shipped 0.95**, **Scrapped 0.05**	
7	**Exit Point**	**Shipped**	Record Final Score: **On**	
8	**Exit Point**	**Scrapped**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Laser cutters	≈ 50%	**50%**	Matches.
Welding robot	80% · 87 s	**80% · 82 s**	Matches: the bottleneck.
Parts in the oven	≈ 6	**5.9** (5.5–6.3)	Little's Law: 0.02 × 300. Measured as the time-averaged count in the oven.
Inspection	60% · ≤ 25 s	**60% · 3 s**	The 25 s assumes random arrivals. The welder and fixed-time oven deliver parts evenly spaced, so inspection rarely queues.
Shipped / scrapped	≈ 540 / 28	**543 / 31**	Matches.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Weld buffer**.
- **Traffic Heatmap**: the welding stage runs hottest.
- **Cost & ROI**: add hourly costs to price option (a) against (b).

COMMON MISTAKES

- Typing the panel's breakdown fields and expecting downtime: the main engine does not apply them yet. Fold availability into the cycle (36 ÷ 0.9 = 40 s).
- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Using a Machine for the oven: it would allow one part at a time and invent a huge queue.

EXPERIMENT

Try: Compare (a) **2** welding robots vs (b) 98% availability (Time **36.7**); then Stress at **+30%**.

Result: Measured: (a) 41% busy, 1 s wait; (b) 72%, 40 s. At +30% the single robot hits 97% with a 383 s wait and rising; two robots hold at 50% and inspection (74%) becomes the next constraint.

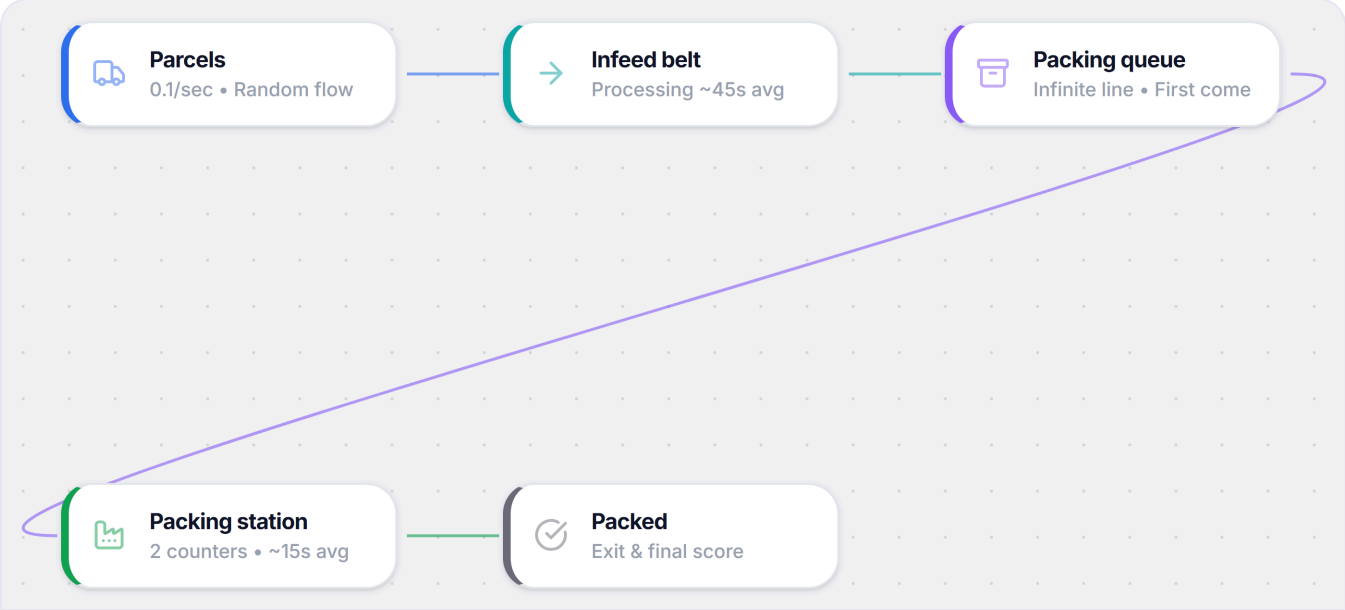
> Downtime is modelled through the cycle time because the main engine does not yet apply the panel's breakdown (MTBF) fields.

GUIDE 13 LOGISTICS EASY

Conveyor Belt to Packing Station

Parcels ride a 30-metre infeed belt (45 seconds) to a packing station with two packers.

QUESTION Can two packers keep up, and what does the belt add?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Parcels → Infeed belt → Packing queue → Packing station → Packed
```

RUN LENGTH	BLOCKS	KEY IDEA
2 hrs	5	A conveyor adds lead time, not capacity loss.

GUIDE 13 LOGISTICS EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Parcels**	**Natural Random Spacing** · Arrival Speed: **0.1**	360 an hour.
2	**Conveyor Step**	**Infeed belt**	Average Duration **45** · **Fixed Constant Duration**	A belt moves every parcel at the same speed and carries many at once.
3	**Waiting Line → Loading Dock**	**Packing queue → Packing station**	Line: defaults · **2** · Time **15** · **Random Service Duration**	Loading Dock is the logistics palette's limited block: use it for any station where only N work at once.
4	**Exit Point**	**Packed**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Parcels	≈ 720	**705** (659–752)	Matches.
Time on the belt	45 s	**45.0 s**	Exact: fixed belt speed.
Parcels on the belt	≈ 4.5	**4.5** (4.2–4.9)	Little's Law: 0.1 × 45. Measured as the time-averaged count on the belt.
Packer utilization	≈ 75%	**72%**	Matches.
Average packing wait	≈ 19 s	**19 s**	Matches.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Packing queue**.
- **Station Counters**: Dock Utilization (here: packers) ≈ 72%.
- **Flow & Balance Check**: parcels still on the belt explain in ≠ out.

COMMON MISTAKES

- Using a Loading Dock for the belt: it would let one parcel ride at a time.
- Skipping the Waiting Line. Without it, waits are still counted on the limited block, but patience (give-ups) cannot work.
- Typing rates per hour or times in minutes. 72 an hour is **0.02**; 4 minutes is **240**.

EXPERIMENT

Try: Slow the belt to **90** s, then try **1** packer.

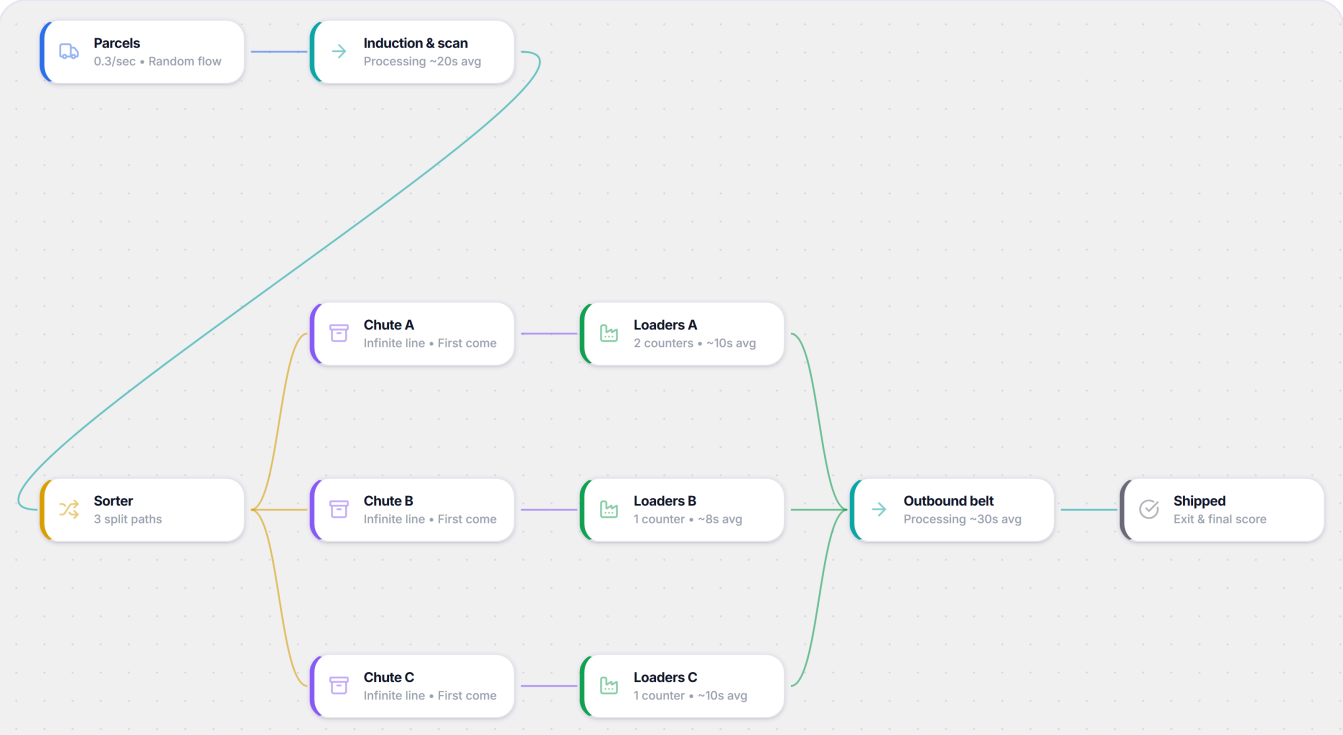
Result: Measured: belt 90 s leaves packers at 76% and 23 s, unchanged. One packer: 99% busy, 20-minute waits, only 468 of 714 packed.

GUIDE 14 LOGISTICS MEDIUM

Parcel Sorter with Three Chutes

A sort hub scans parcels on an induction belt, sorts them to three chutes with their own loading teams, and merges them onto an outbound belt.

QUESTION Is each chute staffed for its share of the volume?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Parcels → Induction & scan → Sorter
Sorter → (0.5) Chute A → Loaders A → Outbound belt
Sorter → (0.3) Chute B → Loaders B → Outbound belt
Sorter → (0.2) Chute C → Loaders C → Outbound belt
Outbound belt → Shipped
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	11	Staffing must follow the sort plan, chute by chute.

GUIDE 14 LOGISTICS MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Parcels**	**Natural Random Spacing** · Arrival Speed: **0.3**	1,080 an hour.
2	**Conveyor Step**	**Induction & scan**	Average Duration **20** · **Fixed Constant Duration**	
3	**Decision Block**	**Sorter**	Paths: **Chute A 0.5**, **Chute B 0.3**, **Chute C 0.2**	
4	**Waiting Line**	**Chute A / Chute B / Chute C**	Three lines, defaults	
5	**Loading Dock**	**Loaders A**	**2** · Time **10** · **Random Service Duration**	
6	**Loading Dock**	**Loaders B**	**1** · Time **8** · **Random Service Duration**	
7	**Loading Dock**	**Loaders C**	**1** · Time **10** · **Random Service Duration**	
8	**Conveyor Step**	**Outbound belt**	Average Duration **30** · **Fixed Constant Duration**	All three docks feed it.
9	**Exit Point**	**Shipped**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Parcels	≈ 1,080	**1,080** (1,027–1,132)	Matches.
Loaders A	75% · 13 s	**73% · 12 s**	Matches.
Loaders B	72% · 21 s	**71% · 16 s**	Matches within spread.
Loaders C	60% · 15 s	**61% · 14 s**	Matches.
Shipped	≈ arrivals	**1,057**	Parcels still on the belts at the end.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Chute B**.
- **Traffic Heatmap**: see which chute runs hottest.
- **Line Buildup Timeline**: compare the three chute lines.

COMMON MISTAKES

- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Three chutes feeding one shared loader block: each chute has its own team.
- Forgetting to connect all three docks to the outbound belt.

EXPERIMENT

Try: Sort plan **0.4 / 0.4 / 0.2**, then give Loaders B a second loader.

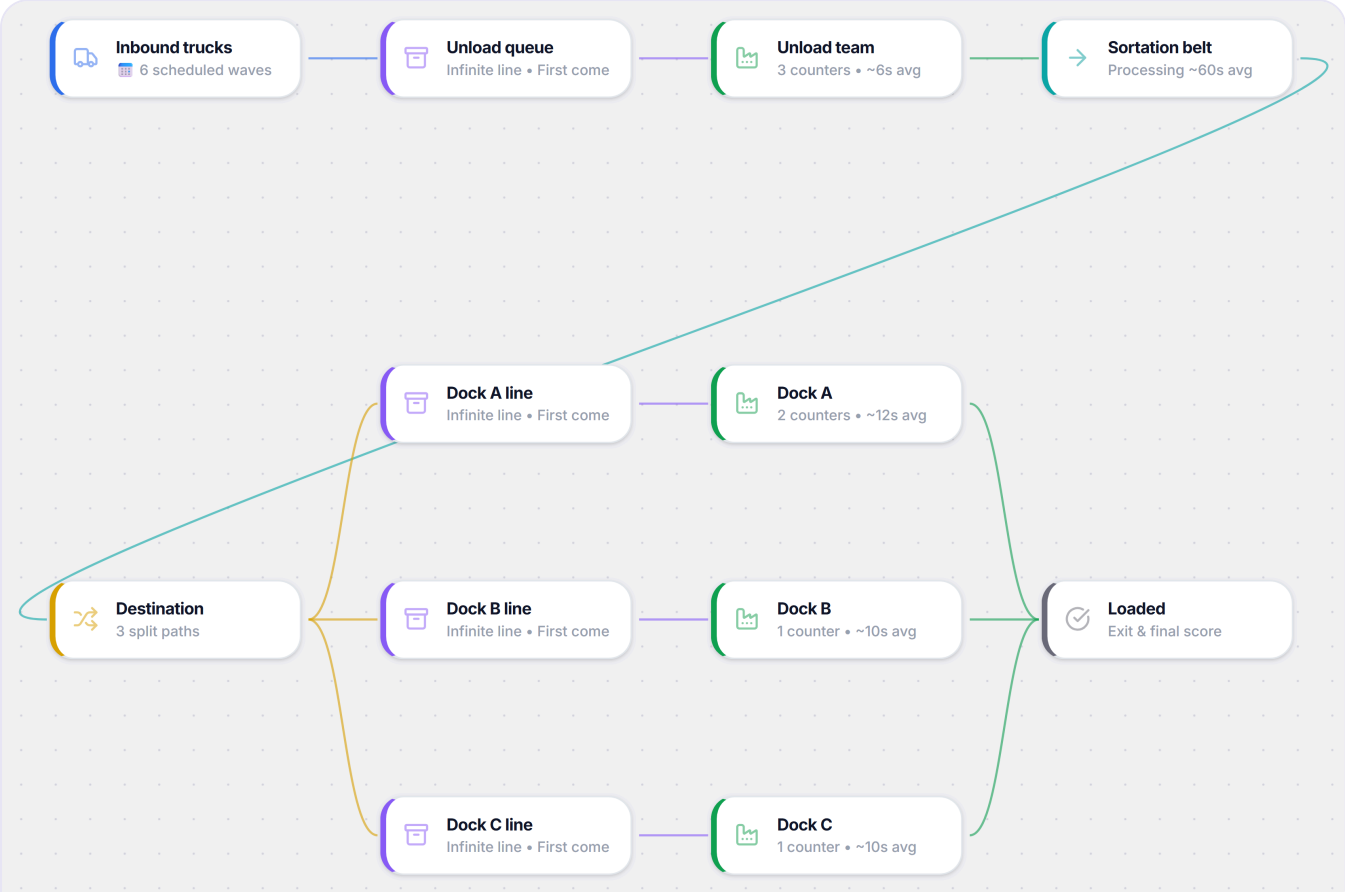
Result: Measured: Loaders B 91% and 114 s (7× longer); with 2 loaders B drops to 46% and 2 s.

GUIDE 15 LOGISTICS ADVANCED

Cross-dock: Truck Waves to Outbound Docks

A truck with 120 parcels arrives every 10 minutes; a 3-person team unloads it onto a sortation belt that feeds three outbound docks.

QUESTION Average utilization looks fine. Where do the waves still create queues?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Inbound trucks → Unload queue → Unload team → Sortation belt → Destination
Destination → (0.4) Dock A line → Dock A → Loaded
Destination → (0.35) Dock B line → Dock B → Loaded
Destination → (0.25) Dock C line → Dock C → Loaded
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	12	Averages hide waves: look at peaks, not means.

GUIDE 15 LOGISTICS ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Start Point**	**Inbound trucks**	**Exact Time Schedule**: Add Arrival six times: at second **0, 600, 1200, 1800, 2400, 3000**, how many **120**	Each entry releases a whole truckload at once.
2	**Waiting Line → Loading Dock**	**Unload queue → Unload team**	Line: defaults · **3** · Time **6** · **Fixed Constant Duration**	Three people, 6 s a parcel: a truck takes 240 s.
3	**Conveyor Step**	**Sortation belt**	Average Duration **60** · **Fixed Constant Duration**	
4	**Decision Block**	**Destination**	Paths: **Dock A line 0.4**, **Dock B line 0.35**, **Dock C line 0.25**	
5	**Waiting Line**	**Dock A / B / C line**	Three lines, defaults	
6	**Loading Dock**	**Dock A**	**2** · Time **12** · **Random Service Duration**	
7	**Loading Dock**	**Dock B**	**1** · Time **10** · **Random Service Duration**	
8	**Loading Dock**	**Dock C**	**1** · Time **10** · **Random Service Duration**	
9	**Exit Point**	**Loaded**	Record Final Score: **On**	All three docks connect here.

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Parcels	720	**720**	Exact: six scheduled trucks.
Unload team · queue wait	40% · 117 s	**40% · 117 s**	Exact: a parcel waits for the ones ahead of it in its truck.
Dock A / B / C utilization	48 / 70 / 50%	**49 / 71 / 50%**	Matches.
Peak line per wave A / B / C	≥ 8 / 18 / 6	**20 / 29 / 16**	The smooth-flow estimate is a floor; random service and routing make each wave bigger.
Loaded	≈ 720	**720**	The last wave clears before the end.

ON THE RESULTS DASHBOARD

- **Executive Summary**: Bottleneck = **Unload queue** because a whole truckload waits at once. That is a wave effect: the team is only 40% busy.
- **Line Buildup Timeline**: the saw-tooth every 10 minutes is the real story.
- **Station Counters**: dock utilizations look comfortable on average.

COMMON MISTAKES

- Using Paced Arrivals for trucks: one Exact Time Schedule entry per truck releases the whole load.
- Decision Block: drawing the arrows is not enough. Open it → Outgoing Paths → **Add Path** for each arrow, type the target block's ID and its chance. Without paths, items stop at the decision and never reach the exit.
- Judging docks on average utilization alone.

EXPERIMENT

Try: Double the unload team to **6**.

Result: Measured: the team drops to 20%, but peak dock lines grow to 35 / 38 / 26 and Dock B's wait rises from 104 s to 155 s. Speeding up one step moves the queue downstream.

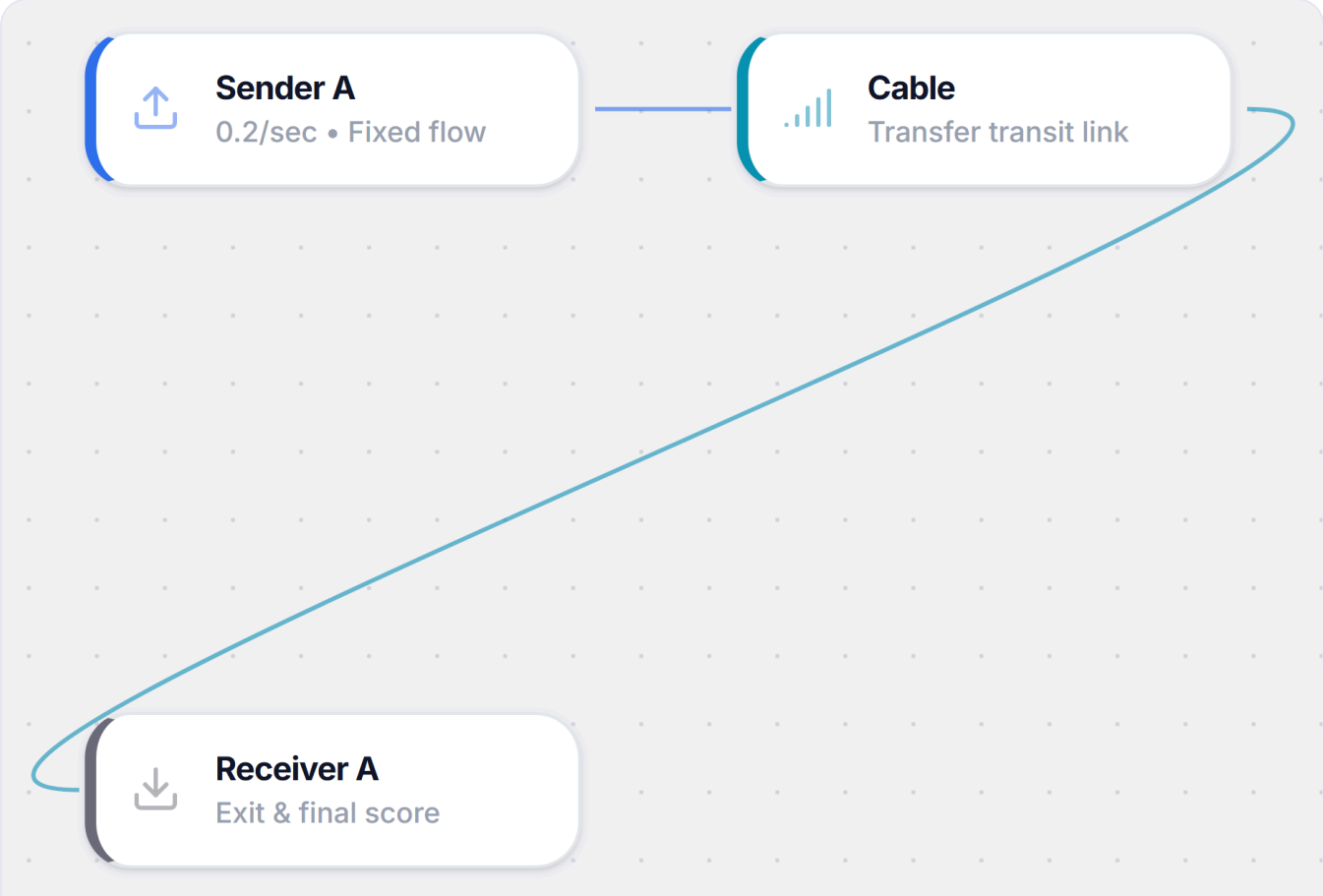
> Watch the queue-depth chart rather than averages: the story here is the saw-tooth every 10 minutes.

GUIDE 16 NETWORK & SIGNAL EASY

Fixed Cable Latency

A sender emits a message every 5 seconds over a link with a fixed 10-second delay.

QUESTION What latency and throughput does the receiver see?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Sender A → Cable → Receiver A
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	3	Latency comes from the link; the sender only changes the count.

GUIDE 16 NETWORK & SIGNAL EASY

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Cable / Event Latency**	New **Network & Signal** project → template gallery → **Cable / Event Latency**	The template presets the channel delay, which the panel does not edit today.
2	**Start Point**	**Sender A**	Check: **Fixed & Exact Spacing** · Arrival Speed **0.2**	Editable.
3	**Transport Channel**	**Cable**	Preset: **10 s** fixed delay, unlimited buffer, late detection on	Carries every message with the same delay.
4	**Exit Point**	**Receiver A**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Messages sent	719	**719**	The message due at exactly 3,600 s is not counted.
Messages received	717	**717**	Two are still on the cable at the end.
Average latency	10	**10.0**	Seconds of simulated time; the chart may label it ms.
Late messages	0	**0**	A fixed delay never exceeds itself.

ON THE RESULTS DASHBOARD

- **Station Counters**: Cable shows 719 in, 717 out.
- **Live Activity Log**: each message is transmitted, then received 10 s later.
- **Flow & Balance Check**: 2 in flight.

COMMON MISTAKES

- Building the channel by hand: a dropped Transport Channel has zero delay and no settings panel. Start from the template.
- Reading the channel's utilization: it reads 100% for any in-flight count.

EXPERIMENT

Try: Set Sender A to **Natural Random Spacing**.

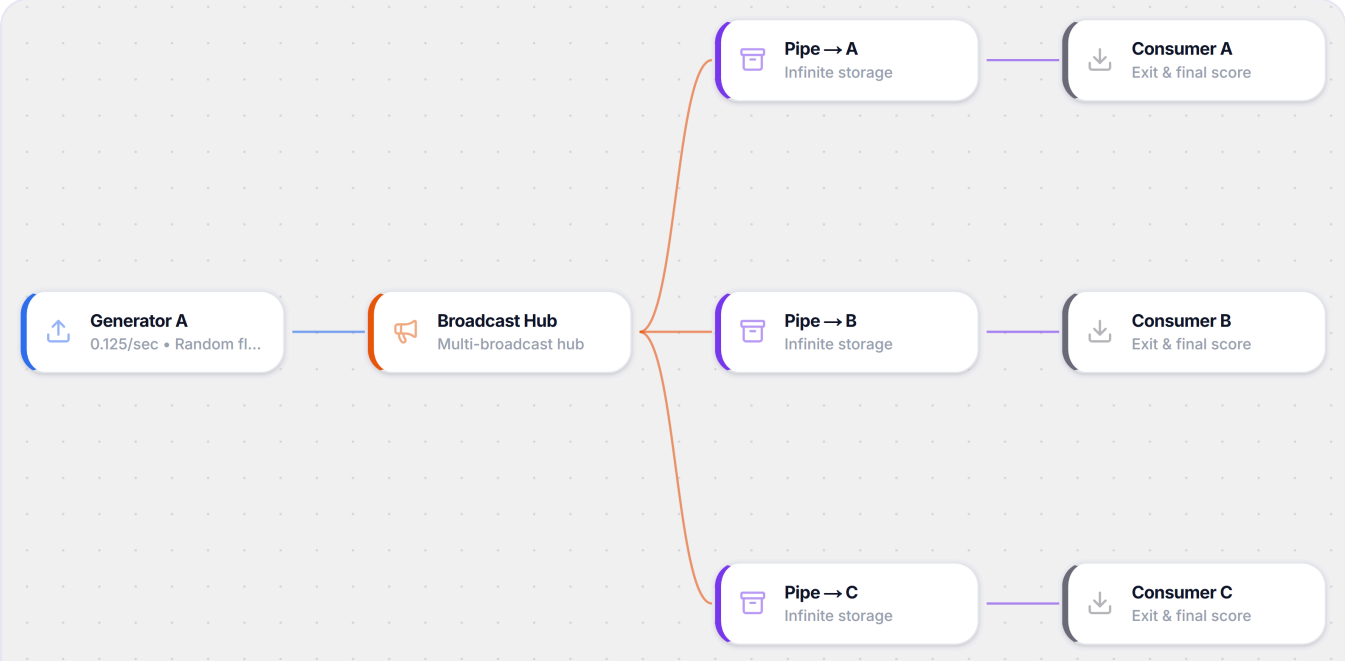
Result: Measured: ≈ 717 sent (694–780), latency still exactly 10.

GUIDE 17 NETWORK & SIGNAL MEDIUM

One Message, Three Subscribers

A publisher sends an event about every 8 seconds; a broadcaster copies it to three subscriber pipes.

QUESTION How much traffic does fan-out create downstream?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
Generator A → Broadcast Hub → Pipe → A / B / C → Consumer A / B / C
```

RUN LENGTH	BLOCKS	KEY IDEA
1 hr	8	Fan-out multiplies load; routing divides it.

GUIDE 17 NETWORK & SIGNAL MEDIUM

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Broadcast Pipe (One-to-Many)**	Network & Signal project → template gallery → **Broadcast Pipe (One-to-Many)**	
2	**Start Point**	**Generator A**	Check: **Even Range Spread** · Arrival Speed **0.125**	Gaps between 4 and 12 s, 8 on average.
3	**Broadcaster**	**Broadcast Hub**	No settings	Copies each message to every connected path.
4	**Storage Rack**	**Pipe → A / B / C**	Preset: unlimited	Hold messages until the consumer takes them.
5	**Exit Point**	**Consumer A / B / C**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Messages generated	≈ 450	**449** (442–461)	Matches.
Each consumer	≈ 450	**449**	Every copy is delivered.
Total completed	≈ 1,350	**1,346**	3× generated: each copy counts at its exit.
Buffer depth	≈ 0	**0**	Consumers take messages immediately.

ON THE RESULTS DASHBOARD

- **Executive Summary**: total completed is 3× the arrivals; that is fan-out, not a bug.
- **Station Counters**: identical counts on all three consumers.
- **Flow & Balance Check**: out = 3 × in.

COMMON MISTAKES

- Reading “completed > arrived” as an error.
- Expecting the Broadcaster to split traffic; it copies.

EXPERIMENT

Try: Replace the Broadcast Hub with a **Decision Block** (paths 0.34 / 0.33 / 0.33).

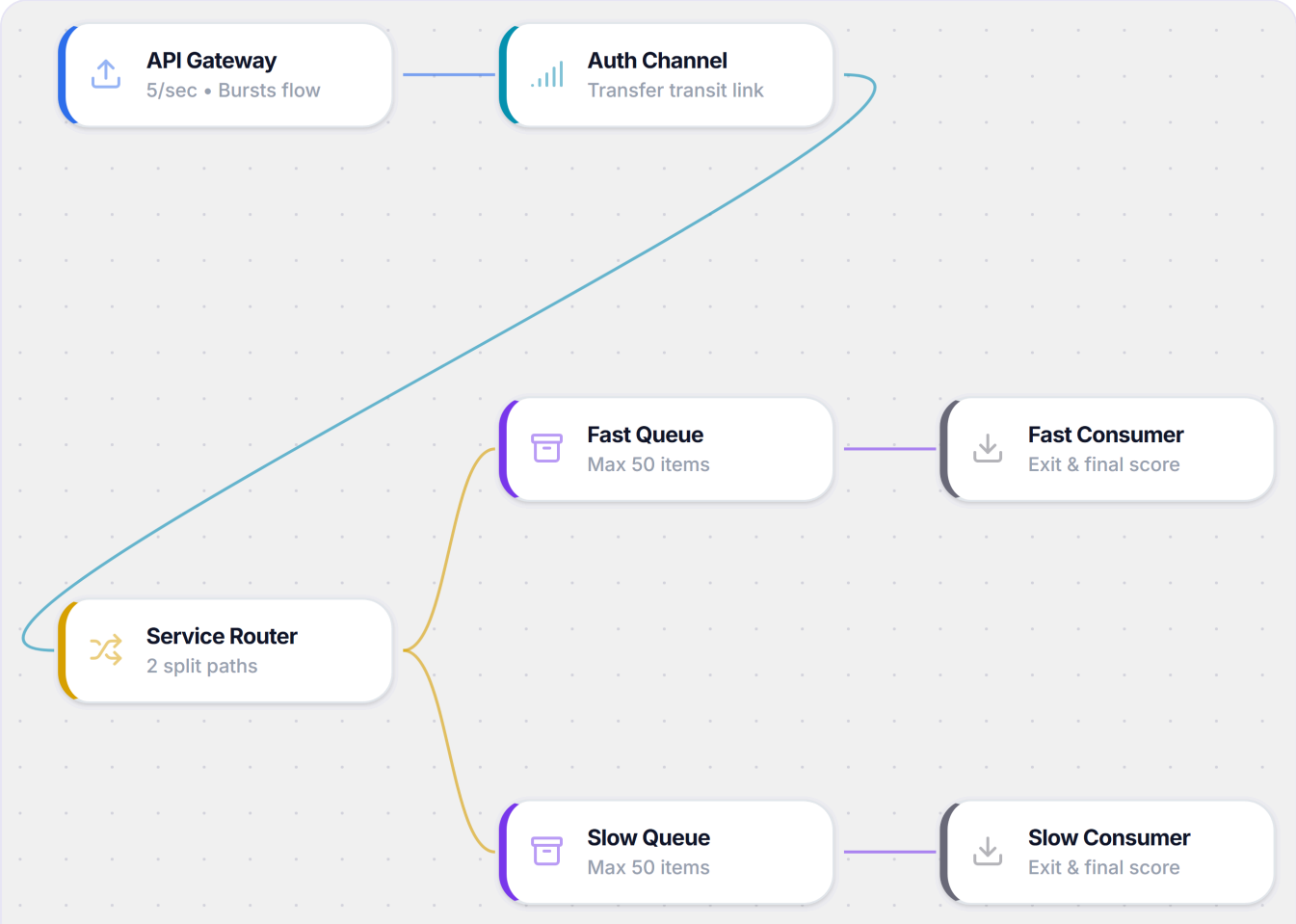
Result: Measured: total completed 446 = generated. Same diagram shape, one third of the load.

GUIDE 18 NETWORK & SIGNAL ADVANCED

Microservice Gateway Under Load

An API gateway forwards about 5 requests a second through an auth link (random delay, 2 s average, 100 in-flight slots) to fast and slow services.

QUESTION At what request rate does the auth link start dropping traffic?



As built on the JustCmul8 editor canvas (light theme). High-resolution: zoom in to read every block.

CONNECTION MAP

```
API Gateway → Auth Channel → Service Router
Service Router → (0.4) Fast Queue → Fast Consumer
Service Router → (0.6) Slow Queue → Slow Consumer
```

RUN LENGTH	BLOCKS	KEY IDEA
10 mins	7	Here the in-flight buffer, not compute, sets the ceiling.

GUIDE 18 NETWORK & SIGNAL ADVANCED

Build & check

#	BLOCK	NAME IT	SETTINGS	WHY
1	**Template**	**Microservice Pipeline**	Network & Signal project → template gallery → **Microservice Pipeline**	
2	**Start Point**	**API Gateway**	Check: **Rush Hour / Burst Spikes** · Arrival Speed **5**	Behaves like random arrivals in the engine.
3	**Transport Channel**	**Auth Channel**	Preset: random delay, mean **2 s**, buffer **100**, late detection on	100 messages can be in flight; more are dropped.
4	**Decision Block**	**Service Router**	Preset paths: **Fast 0.4**, **Slow 0.6**	
5	**Storage Rack**	**Fast Queue / Slow Queue**	Preset	
6	**Exit Point**	**Fast / Slow Consumer**	Record Final Score: **On**	

CHECK THE RESULTS	THEORY	MEASURED	WHY IT MATCHES, OR WHY NOT
Requests	≈ 3,000	**3,017** (2,938–3,083)	Matches.
Average auth latency	≈ 2	**1.99**	Matches.
Late messages	≈ 37%	**36.6%** (1,105)	e^{-1}: an exponential delay exceeds its mean 36.8% of the time.
Dropped	≈ 0	**0**	≈ 10 in flight vs 100 slots.
Fast / Slow	≈ 1,200 / 1,800	**1,209 / 1,799**	Matches the 0.4 / 0.6 split.

ON THE RESULTS DASHBOARD

- **Station Counters**: Auth Channel shows late and dropped counts.
- **Line Buildup Timeline**: Buffer Depth stays far below 100 at 5 per second.
- **Executive Summary**: Throughput ≈ 5 per second.

COMMON MISTAKES

- Treating “late” as failure: it counts messages slower than the nominal delay, so ≈ 37% is normal for random delays.
- Raising the request rate in big jumps; step it up to find where drops begin.

EXPERIMENT

Try: Raise Arrival Speed to **40**, **50**, then **60**.

Result: Theory (Erlang loss): 0.4% / 7.6% / 19.6%. **Measured: 0.34% / 7.8% / 19.4% dropped.**

B

Pitch catalogue

Thirty places where discrete-event simulation earns its keep, five per domain.

About these use cases

They are typical industry applications of this kind of simulation and what JustCmul8 can model for each. They are not customer case studies, and no company named or implied here is a JustCmul8 customer.

B PITCH CATALOGUE

People & Service

Palette: Start Point · Waiting Line · Limited Counter · Counter · Decision Block · Exit Point

SIMPLE Bank branch teller staffing

Banking

BUSINESS PROBLEM
Lines at lunch and month-end; customers leave.

WHAT YOU MODEL
Start Point → Waiting Line (patience) → Limited Counter

KPIS → DECISION
Wait time, teller utilization, walk-aways → **Tellers per time band**

SIMPLE Café or quick-service counter

Food service

BUSINESS PROBLEM
Morning rush overwhelms the counter.

WHAT YOU MODEL
Start Point → Waiting Line → Limited Counter → Exit Point

KPIS → DECISION
Wait, queue length, throughput → **Staff on the rush shift**

SIMPLE Metro and railway ticketing

Transit

BUSINESS PROBLEM
Token queues back up into the concourse at peak.

WHAT YOU MODEL
Decision Block for card vs token, two counters in series

KPIS → DECISION
Wait, walk-aways, gate load → **Counters and gates to open**

COMPLEX Hospital OPD and emergency triage

Healthcare

BUSINESS PROBLEM
Long waits, uneven doctor load, pharmacy pile-ups.

WHAT YOU MODEL
Registration → Triage (Decision) → parallel doctor pools → pharmacy

KPIS → DECISION
Wait per stage, utilization, patients leaving → **Doctor rostering, triage rules**

COMPLEX Contact centre with escalation

BPO / support

BUSINESS PROBLEM
Callers abandon; L2 agents idle or swamped.

WHAT YOU MODEL
Waiting Line (patience) → L1 agents → Decision → L2 agents

KPIS → DECISION
Abandon rate, service level, utilization → **Agents per tier and escalation rate**

Traffic & Vehicles

Palette: Start Point · Waiting Line · Service Bay (Limited) · Processing Block · Decision Block · Exit Point

SIMPLE Fuel and CNG station pumps

Retail fuel

BUSINESS PROBLEM
Queues spill onto the road at peak.

WHAT YOU MODEL
Start Point → Waiting Line (patience) → Service Bay

KPIS → DECISION
Wait, drive-aways, pump utilization → **Number of pumps or dispensers**

SIMPLE Toll plaza lane mix

Highways

BUSINESS PROBLEM
Cash lanes back up while tag lanes idle.

WHAT YOU MODEL
Decision Block by payment type → lane pools → Processing Block

KPIS → DECISION
Wait per lane type, utilization → **Lanes per payment type**

COMPLEX EV charging hub sizing

E-mobility

BUSINESS PROBLEM
Chargers either idle or fully booked with long waits.

WHAT YOU MODEL
Decision fast vs slow charge → two Service Bay pools

KPIS → DECISION
Wait, charger utilization, drivers turned away → **Charger count and fast/slow mix**

COMPLEX Car service workshop

Automotive aftersales

BUSINESS PROBLEM
Bays congested; delivery promises missed.

WHAT YOU MODEL
Advisors → Decision (minor/major) → bay pools → wash

KPIS → DECISION
Lead time, bay utilization → **Bays, technicians, booking caps**

COMPLEX Port or terminal truck gate

Ports and logistics parks

BUSINESS PROBLEM
Truck queues outside the gate; demurrage.

WHAT YOU MODEL
Gate lanes → Decision (inspection) → yard Processing Block

KPIS → DECISION
Truck turnaround, queue length → **Gate lanes and inspection rate**

B PITCH CATALOGUE

Liquid & Material

Palette: Start Point · Storage Tank · Processing Unit · Valve / Splitter · Exit Point

SIMPLE Storm-water retention sizing

Municipal water

BUSINESS PROBLEM
Will the tank overflow in a design storm?

WHAT YOU MODEL
Start Point (pulses) → Storage Tank → Exit Point

KPIS → DECISION
Time to full, overflowed volume → **Tank size and inflow limits**

SIMPLE Tanker unloading at a depot

Fuel and chemicals

BUSINESS PROBLEM
Receiving tanks fill before the next dispatch.

WHAT YOU MODEL
Exact Time Schedule of tankers → Storage Tank

KPIS → DECISION
Fill time, rejected loads → **Tank capacity and delivery schedule**

SIMPLE Water treatment train

Utilities

BUSINESS PROBLEM
Filters and clear-water tanks mismatched in size.

WHAT YOU MODEL
Valve / Splitter → tanks → Processing Units

KPIS → DECISION
Tank levels, treated volume → **Split ratio and tank sizing**

COMPLEX Dairy or beverage batch processing

Food and beverage

BUSINESS PROBLEM
Pasteuriser and storage tanks limit daily output.

WHAT YOU MODEL
Processing Unit (batch time) → Valve (QA) → product tanks

KPIS → DECISION
Batches in process, tank fill time → **Vessel count and release schedule**

COMPLEX Chemical batch reactor scheduling

Chemicals and pharma

BUSINESS PROBLEM
How many reactors a new schedule needs.

WHAT YOU MODEL
Fixed-spacing releases → Processing Unit → QC split → tanks

KPIS → DECISION
Reactors needed (Little's Law), off-spec volume → **Capex for reactors and tanks**

Manufacturing

Palette: Start Point · Waiting Line · Machine / Robot · Assembly Step · Decision Block · Exit Point

SIMPLE Single machine cell

Discrete manufacturing

BUSINESS PROBLEM
Is the CNC overloaded? How much WIP is normal?

WHAT YOU MODEL
Start Point → Waiting Line → Machine / Robot

KPIS → DECISION
Utilization, wait, WIP → **Shift pattern, second machine**

SIMPLE Paint or curing conveyor oven

Automotive, appliances

BUSINESS PROBLEM
How many parts must the oven hold?

WHAT YOU MODEL
Assembly Step as the oven (fixed time, many at once)

KPIS → DECISION
Parts inside (Little's Law), lead time → **Oven length and conveyor speed**

COMPLEX QC and rework loop

Precision engineering

BUSINESS PROBLEM
Rework silently consumes machine capacity.

WHAT YOU MODEL
Machine → Decision (QC) → Assembly Step (rework) → back

KPIS → DECISION
Rework passes, machine load, wait → **Quality investment vs extra capacity**

COMPLEX Assembly line balancing

Electronics, automotive

BUSINESS PROBLEM
Uneven stations starve or block each other.

WHAT YOU MODEL
Chain of Machine / Robot stations with buffers

KPIS → DECISION
Station utilization, bottleneck, throughput → **Re-balancing tasks between stations**

COMPLEX Bottleneck and downtime (OEE)

Any line with robots

BUSINESS PROBLEM
Second robot or better maintenance?

WHAT YOU MODEL
Availability folded into cycle time; Compare scenarios

KPIS → DECISION
Throughput ceiling, wait, utilization → **Capex vs maintenance programme**

Logistics

Palette: Start Point · Waiting Line · Loading Dock · Conveyor Step · Decision Block · Exit Point

SIMPLE Packing station staffing

E-commerce fulfilment

BUSINESS PROBLEM
Packers cannot keep up with the infeed belt.

WHAT YOU MODEL
Conveyor Step → Waiting Line → Loading Dock

KPIS → DECISION
Packer utilization, wait, belt load → **Packers per shift**

SIMPLE Inbound dock scheduling

Warehousing

BUSINESS PROBLEM
Trucks wait for a free dock.

WHAT YOU MODEL
Exact Time Schedule → Waiting Line → Loading Dock

KPIS → DECISION
Truck wait, dock utilization → **Number of docks and slot booking**

SIMPLE Returns processing

Retail and e-commerce

BUSINESS PROBLEM
Returns pile up; refunds are late.

WHAT YOU MODEL
Inspection → Decision (restock / refurbish / scrap)

KPIS → DECISION
Backlog, lead time → **Inspectors and routing rules**

COMPLEX Parcel sortation hub

Courier and postal

BUSINESS PROBLEM
Chute staffing does not match the sort plan.

WHAT YOU MODEL
Conveyor Steps → Decision (sorter) → chute docks → outbound belt

KPIS → DECISION
Chute wait, loader utilization → **Loaders per chute, sort plan changes**

COMPLEX Cross-dock wave planning

3PL and retail distribution

BUSINESS PROBLEM
Truck waves create temporary queues at outbound docks.

WHAT YOU MODEL
Scheduled truckloads → unload team → belt → docks

KPIS → DECISION
Peak queue per wave, dock utilization → **Unload team size and truck intervals**

Network & Signal

Palette: Start Point · Transport Channel · Storage Rack · Broadcaster · Decision Block · Exit Point

SIMPLE Link latency budgeting

Telecom and IoT

BUSINESS PROBLEM
Does end-to-end latency meet the SLA?

WHAT YOU MODEL
Start Point → Transport Channel → Exit Point (template)

KPIS → DECISION
Latency, late messages → **Link choice and SLA targets**

SIMPLE Pub/sub fan-out sizing

Event-driven software

BUSINESS PROBLEM
Subscribers multiply downstream traffic.

WHAT YOU MODEL
Start Point → Broadcaster → Storage Racks → consumers

KPIS → DECISION
Messages per consumer, total load → **Consumer capacity and topic design**

COMPLEX API gateway buffer sizing

SaaS and fintech

BUSINESS PROBLEM
Requests dropped at peak.

WHAT YOU MODEL
Gateway → Transport Channel (buffer) → Decision → services

KPIS → DECISION
Drop rate, latency, late % → **Buffer size and rate limits**

COMPLEX IoT telemetry ingestion

Industrial IoT, smart meters

BUSINESS PROBLEM
Burst reporting overwhelms the ingest link.

WHAT YOU MODEL
Burst arrivals → Transport Channel → Storage Rack → processors

KPIS → DECISION
Drops, buffer depth → **Buffer, batching and reporting interval**

COMPLEX Transaction routing (fast and slow paths)

Payments

BUSINESS PROBLEM
Slow path congestion delays settlement.

WHAT YOU MODEL
Decision Block routing → separate processing paths

KPIS → DECISION
Latency per path, throughput → **Routing rules and path capacity**